

MASTER PROJECT

A journey around asymptotically rigid mapping class groups and their cube complexes

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Abstract

The present master project intends to explore the asymptotically rigid mapping class groups of arboreal surfaces and their cube complexes. This notion appears in the study of Thompson's groups when we study them from the perspective of mapping class groups of infinite surfaces.

A general method to study the algebraic structure of a given group consists in building a combinatorial complex on which it acts and then studying the geometry of this complex. If the latter cube complex has particular characteristics one can obtain interesting results about the group acting on it. The first part of the project aims to find a CAT(0) cube complex on which the asymptotically rigid mapping class group of an infinitely-punctured surface obtained by thickening planar trees acts. To this end we start from the construction presented in Section 3 in [GLU22]. We deduce some properties of this cube complex and we show that it is the desired cube complex for asymptotically rigid mapping class groups associated to a certain family of planar trees. Then, we modify a bit this construction to obtain a CAT(0) cube complex associated to a larger family of planar trees. Finally, by taking our construction and Theorem 1.4 in [Hag07] we deduce geometric properties on the asymptotically rigid mapping class groups from the fact that it acts on a CAT(0) cube complex.

The last goal of this thesis is to compute a presentation for the asymptotically rigid mapping class group associated with a certain family of trees. The main idea is to use the algorithm of Kenneth Brown presented in [Bro84]. In the latter article, Brown gives a method, derived from Bass-Serre theory, to compute the presentation of a group acting on a connected CW-complex.

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1 Introduction

Thompson's groups F, T and V were introduced by Richard Thompson in some unpublished handwritten notes in 1965. They give a fruitful source of counter-examples in group theory [CFP96]. Furthermore, they also consist of a source of inspiration to build other groups by varying their construction. The braided Thompson's groups are extensions of Thompson's groups by infinite braid groups. The latter groups can be recovered as mapping class groups of infinite surfaces [FKS11] giving rise to the notion of asymptotically rigid mapping class groups.

To motivate the definition of *asymptotically rigid mapping class groups*, let us first roughly introduce Thompson's groups F, T, V. They form a chain of subgroups $F \subset T \subset V$ of a larger group $\mathrm{AIsom}(A)$, which consists of (equivalence classes of) quasi-automorphisms of a rooted infinite binary tree, i.e. permutations of the vertices that are almost everywhere graph automorphisms (i.e. only finitely many vertices are permuted without necessarily preserving adjacency). The equivalence relation identifies quasi-automorphisms that only differ by a permutation of finitely many vertices. Adding adequate restrictions on the almost automorphisms yields the three Thompson's groups F, T, V.

The path from quasi-automorphisms to asymptotically rigid mapping class groups is parallel to the path from permutation groups to braid groups. Elements of the latter not only permute vertices, but they also remember how the vertices move around each other. Doing so requires embedding the vertices in a plane or another surface, then elements of the braid groups can be identified with (isotopy classes of) homeomorphisms of the surface preserving the set of vertices. Applying the same line of thought to Thompson's group $\mathrm{AIsom}(A)$ requires embedding the infinite binary tree in a surface, called an *arboreal surface*, and then considering homeomorphisms of this surface preserving the set of vertices. However, to correctly translate the condition that the permutations are almost tree automorphisms, we need to endow the arboreal surface with a rigid structure that reflects adjacency of vertices. Then asking that the surface homeomorphisms asymptotically preserve this rigid structure is the correct translation of the notion of almost automorphisms.

We start by fixing a locally finite planar tree A embedded into the plane. The associated arboreal surface $\mathscr{S}(A)$ is the surface with boundary obtained by thickening A in the plane. The corresponding rigid structure is defined as follows: first decompose $\mathscr{S}(A)$ into polygons so that each polygon contains exactly one vertex of the underlying tree in its interior. The polygons are defined by a family of pairwise nonintersecting arcs whose endpoints are on the boundary of $\mathscr{S}(A)$ and such that each arc crosses once and transversely a unique edge of the underlying tree A. Hence we obtain the rigid surface $\mathscr{S}^\sharp(A)$. See Figure 1 for an illustration.

A homeomorphism of $\mathscr{S}^\sharp(A)$ is called *asymptotically rigid* if it sends all but finitely many polygons of the rigid structure to polygons. The group of isotopy classes of orientation-preserving asymptotically rigid homeomorphisms is denoted by $\mathfrak{mod}(A)$ and is called *the asymptotically rigid mapping class group* associated to A. For instance, if B denotes the infinite binary tree then $\mathfrak{mod}(B)$ is a braided extension of Thompson's group T.

In this project we study the asymptotically rigid mapping class group associated to the trees $A_{n,m}$ having a vertex of valence m while all the other vertices have valence $n+1$. This group is a braided version of the Higman-Thompson groups. A good way to understand this group is to study geometric objects on which the group acts. To this end, we introduce

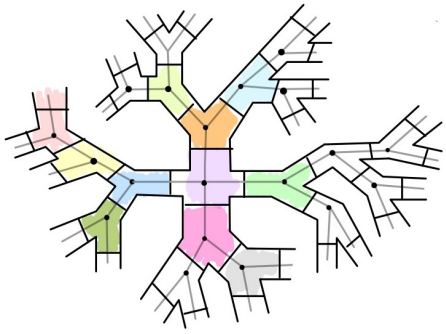


Figure 1

in Section 2 some background about cube complexes and median graphs. This gives us a toolbox to build a suitable cube complex on which $\mathfrak{mod}(A_{n,m})$ acts. In particular, we state Proposition 2.28 giving the CAT(0)ness of a cube complex under some conditions. This proposition follows from Theorem 3.3.1 in [Gen22] (which gives conditions to obtain a median graph) and Theorem 6.1 in [Che00]. The latter states that cube completion of median graphs and cube complexes are the same. We give details of the proof of one direction.

In the third section we study the cube complex $\mathscr{C}(A)$ introduced in [GLU22]. The vertices are pairs $[\Sigma, \phi]$ modulo a certain equivalence relation. Here Σ is a subsurface of $\mathscr{S}^\sharp(A)$ which consists of a union of finitely many polygons and ϕ is an asymptotically rigid homeomorphism (Definition 3.1). By using Proposition 2.28, we study when $\mathscr{C}(A_{n,m})$ is CAT(0) or not. In particular we have to study 3-corners in $\mathscr{C}(A_{n,m})$. This study leads us to our main result:

Theorem 3.25. Let $n \geq 1$, then the cube complex $\mathscr{C}(A_{n,m})$ is CAT(0) if and only if $1 \leq m \leq n+1$.

In [GLU22] the authors extract from $\mathscr{C}(A_{n,m})$ a suitable subcomplex $\mathscr{SC}(A_{n,m})$ called the spine. We can extend the previous result to it. Now, to obtain a cube complex associated to $\mathfrak{mod}(A_{n,m})$ that is CAT(0) for a larger family of trees we modify in Section 4 the rigid structure of $\mathscr{S}^\sharp(A)$ by pushing the puncture of the central polygon outside. Hence we obtain a new cube complex associated to $\mathfrak{mod}(A_{n,m})$ and called $\mathscr{D}(A_{n,m})$. The latter is simply-connected, the proof is very similar to the one for $\mathscr{C}(A_{n,m})$ presented in [GLU22]. By using again Proposition 2.28, we study when $\mathscr{D}(A_{n,m})$ is CAT(0) to obtain:

Theorem 4.5. Let $m \geq 3$ and $n \geq 1$, then the cube complex $\mathscr{D}(A_{n,m})$ is CAT(0).

In the last section, the goal is to prove the finite presentability of $\mathfrak{mod}(A_{1,m})$. To do this we first introduce the height-bounded complex $\mathscr{SC}^m(A_{1,m})$ which is a subcomplex of the spine. We prove that the height-bounded complex is simply connected by studying its descending link as in Section 5 in [GLU22].

Proposition 5.10. For $m \geq 3$, the cube complex $\mathscr{SC}^m(A_{1,m})$ is simply connected.

By studying the action of $\mathfrak{mod}(A_{1,3})$ on its height-bounded complex and using the algorithm introduced by Kenneth Brown in [Bro84] we compute a finite presentation for $\mathfrak{mod}(A_{1,3})$ in Theorem 5.14. Then, we give a proof of its finite presentability by using Theorem 4 in [Bro84]:

Proposition 5.3. Let $m \geq 3$, then $\mathfrak{mod}(A_{1,m})$ is finitely presented.

2 Preliminaries

2.1 Some words about Thompson's groups

Thompson's groups are three groups, $F \subseteq T \subseteq V$ with unusual properties, which provide a fruitful source of counterexamples. The Higman-Thompson groups of type (m, n) were introduced by Graham Higman and are denoted $G_{n,r}$. They provide a generalisation of Thompson's groups. Here we introduce briefly the definitions of the previous groups presented in [Gen].

Take a tree A, the group of quasi-automorphisms of A, $\mathrm{QAut}(A)$ consists of bijections $A^0 \longrightarrow A^0$ preserving the adjacency and non-adjacency for all but finitely many pairs of vertices. The subgroup of finitely supported bijections $A^0 \longrightarrow A^0$, where A^0 is the 0-skeleton, is a normal subgroup of $\mathrm{QAut}(A)$, we denote it $\mathrm{FBij}(A)$.

The group of almost-isometries $\mathrm{AIsom}(A)$ is the quotient group $\mathrm{QAut}(A)/\mathrm{FBij}(A)$. An almost-isometry is also given by a triple (B, f, C), where B and C are finite subtrees of A and $f : A - B \longrightarrow A - C$ is an isometry. We define an equivalence relations on the set of almost-isometries. Two almost-isometries (B, f, C) and (B', f', C') are equivalent if and only if f and g agree on some cofinite subset. Consider Figure 2 which represents an almost-isometry (B, f, C). Here A is the infinite binary tree, B is the red tree on the left of the picture and C is the one on the right. Then f is an isometry from $A - B$ to $A - C$ so the components can be permutated and one has an isometry on each component.

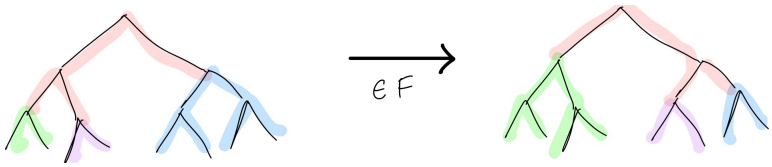


Figure 2

We obtain Thompson's groups by taking the infinite binary tree and imposing conditions on the permutation of the components and the isometries on components.

The group F corresponds to almost isometries having permutations and isometries that preserve the left-right order induced by the plane. See Figure 2.

The group T, corresponds to almost isometries having permutations preserving the cyclic order on the components and isometries that preserve the left-right order on each component. See Figure 3 at the top.

The group V, corresponds to almost isometries having any permutations on the components and isometries that preserve the left-right order on each component. See the middle element in Figure 3.

Let $A_{n,m}$ be the tree with one vertex of valence m and all the others of valence $n + 1$. In the definition of V we can replace the infinite binary tree with $A_{n,m}$ More precisely we define *Higman-Thompson group* of type (n, m) as $V_{n,r} := \mathrm{AIsom}(A_{n,m})$.

2.2 From Thompson's groups to asymptotically rigid mapping class group

In [FKS11], the authors introduce the braided Thompson groups, which are extensions of Thompson's groups by infinite braid groups. In this section we present the framework used

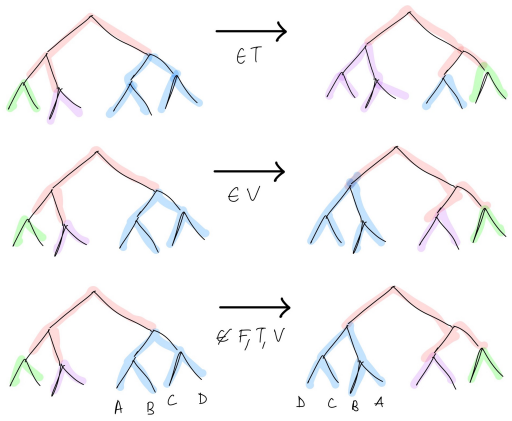


Figure 3

by the authors which is also introduced in[GLU22].

Let A be a locally finite planar tree, its *arboreal surface* is the surface $\mathscr{S}(A)$ obtained by embedding and thickening A into the plane. Then, let $\mathscr{S}^\sharp(A)$ be the surface obtained from $\mathscr{S}(A)$ by adding a puncture for each vertex of the underlying tree A.

We give a rigid structure to $\mathscr{S}^\sharp(A)$ by dividing the surface into polygons with a family of pairwise non-intersecting arcs whose endpoints are on the boundary of the surface such that:

- each arc crosses once and transversely a unique edge of the underlying tree A,
- each polygon contains exactly one puncture in its interior.

See Figure 4 for an illustration when $A_{2,4}$ is the tree having one vertex of valence four while the others have valence two.

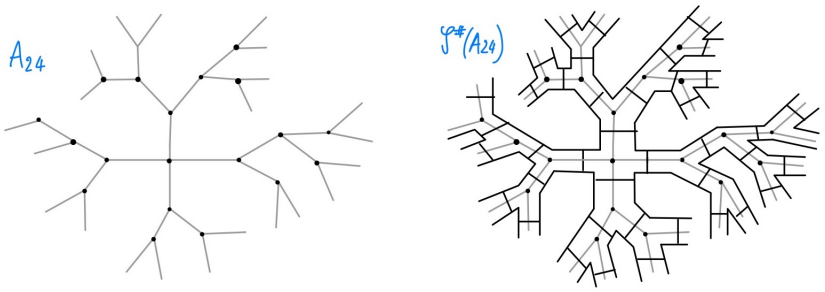


Figure 4

Definition 2.1. Let A be a locally finite tree, a subsurface $\Sigma \subseteq \mathscr{S}^\sharp(A)$ is said to be *admissible* if it can be written as a finite union of polygons from the rigid structure.

The *heigth* of Σ, denoted $h(\Sigma)$, is the number of punctures inside Σ.

The *frontier* of Σ is the union of all the arcs of the rigid structure, called the *frontier-arcs*, lying in the boundary of Σ, it is denoted $\mathrm{Fr}(\Sigma)$. A polygon H is *adjacent* to Σ if it shares an arc with the frontier of Σ.

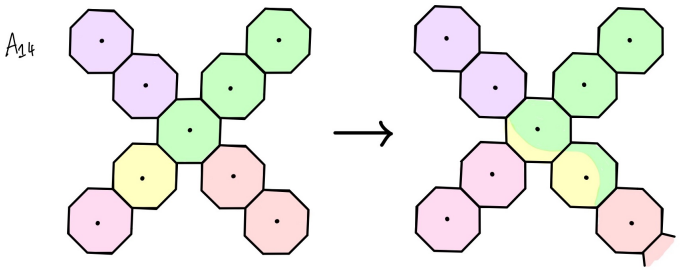


Figure 5

Definition 2.2. Let A be a locally finite tree. An *asymptotically rigid homeomorphism* of $\mathscr{S}^\sharp(A)$ is an element ϕ from $\mathscr{S}^\sharp(A)$ to itself which respects the rigid structure almost everywhere. That is, there exists an admissible surface Σ, called the *support* of ϕ, such that:

- its image $\phi(\Sigma)$ is an admissible surface,
- outside Σ, the map ϕ sends polygons to polygons, i.e. $\phi : \mathscr{S}^\sharp(A) - \Sigma \longrightarrow \mathscr{S}^\sharp(A) - \phi(\Sigma)$ is *rigid*.

See Figure 5 for an example of an asymptotically rigid homeomorphism. We call $\mathfrak{mod}(A)$ the group of isotopy classes of orientation-preserving homeomorphisms of $\mathscr{S}^\sharp(A)$.

Let B be the infinite unrooted binary tree. We take the surface $\mathscr{S}^\sharp(B)$, see Figure 6. Then, Thompson group T is a braided extension of $\mathfrak{mod}(B)$. That is, there exists an exact

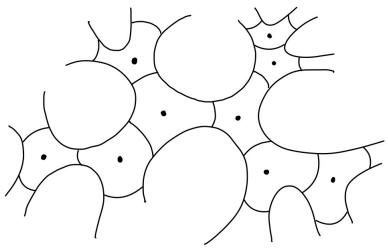


Figure 6: from [FKS11]

sequence

$$1 \longrightarrow \mathcal{B}_\infty \longrightarrow \mathfrak{mod}(B) \longrightarrow T \longrightarrow 1$$

given by forgetting the punctures, see Section 5 in [FKS11].

By forgetting the punctures of $\mathscr{S}^\sharp(A_{n,m})$ we obtain a forgetful map from $\mathfrak{mod}(A_{n,m})$ to $\mathrm{Mod}(\mathscr{S}(A_{n,m}))$. Let $\mathrm{Mod}_f(\mathscr{S}(A_{n,m}))$ be the image of $\mathfrak{mod}(A_{n,m})$ under this map which coincides with the Higman–Thompson group $V_{n,m}$. Then the latter is a braided extension of $\mathfrak{mod}(A_{n,m})$:

$$1 \longrightarrow \mathcal{B}_\infty \longrightarrow \mathfrak{mod}(A_{n,m}) \longrightarrow \mathrm{Mod}_f(\mathscr{S}(A_{n,m})) \longrightarrow 1.$$

Now we consider the tree $A_{1,m}$. The Houghton group H_n is the finite index subgroup in $\mathrm{QAut}(A_{1,m})$ which consists of elements stabilising each strand at infinity of $A_{1,m}$. Then the subgroup of $\mathfrak{mod}(A_{1,m})$ stabilising each strand at infinity of $\mathscr{S}^\sharp(A_{1,m})$ has finite index and

is called the braided Houghton group, it is denoted by $\operatorname{br} H_n$. The group $\operatorname{br} H_n$ is a braided extension of H_n:

$$1 \longrightarrow \mathcal{PB}_\infty \longrightarrow \operatorname{br} H_n \longrightarrow H_n \longrightarrow 1.$$

2.3 Other definitions of Thompson's groups

In this section we provide other definitions of Houghton groups and Thompson's groups that involve the notion of dyadic rearrangements.

Definition 2.3. A *dyadic subdivision* of the unit interval $\mathrm{I} = [0, 1]$, is a subdivision of I into smaller intervals obtained by cutting it repeatedly in halves. For instance,

- $[0, 1] = [0, 1]$ (no cuts),
- $[0, 1] = [0, \frac{1}{4}] \cup [\frac{1}{4}, \frac{1}{2}] \cup [\frac{1}{2}, 1]$, (two cuts).

To a dyadic subdivision one can associate a rooted binary tree (rooted trees where every non-leaf vertex have two children) as follows:

1. Start with a root, $\bullet$,
2. The first cut is always cutting $[0, 1]$ at $\frac{1}{2}$.
3. Now, each additional cut is represented by adding two legs to the cut interval.

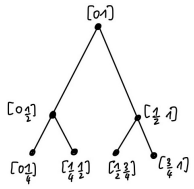


Figure 7

For example, $[0, 1] = [0, \frac{1}{4}] \cup [\frac{1}{4}, \frac{1}{2}] \cup [\frac{1}{2}, \frac{3}{4}] \cup [\frac{3}{4}, 1]$ corresponds to Figure 7. Thus, each leaf of the binary tree represents an interval of the corresponding dyadic subdivision, and each internal vertex represents an interval appearing in the cutting process.

Definition 2.4. A *dyadic rearrangement* of the unit interval is a piecewise linear homeomorphism f of $[0, 1]$ to itself, sending linearly via affine maps a dyadic subdivision to another having the same number of cuts. Note that the starting and ending dyadic subdivision automatically have the same number of leaves.

Dyadic rearrangements of the unit interval form a group F.

For example, the map

$$f\colon [0,1] \to [0,1]\colon x \mapsto \begin{cases} x & \text{if } 0 \le x \le \frac{1}{4}, \\ \frac{x}{2} + \frac{1}{8} & \text{if } \frac{1}{4} \le x \le \frac{1}{2}, \\ \frac{x}{2} + \frac{1}{8} & \text{if } \frac{1}{2} \le x \le \frac{3}{4}, \\ 2x - 1 & \text{if } \frac{3}{4} \le x \le 1 \end{cases},$$

is a dyadic rearrangement starting with the subdivision $[0, \frac{1}{4}] \cup [\frac{1}{4}, \frac{1}{2}] \cup [\frac{1}{2}, \frac{3}{4}] \cup [\frac{3}{4}, 1]$ and ending with the subdivision $[0, \frac{1}{4}] \cup [\frac{1}{4}, \frac{3}{8}] \cup [\frac{3}{8}, \frac{1}{2}] \cup [\frac{1}{2}, 1]$.

A dyadic rearrangement is entirely determined by the pair of dyadic subdivisions it relates. We depict such a pair as opposing trees, with the starting one above and the ending one below, so that the corresponding sub-intervals are facing each other. For example, the above dyadic rearrangement f is represented by the pair of opposing trees on the right in Figure 9. Furthermore, different pairs of dyadic subdivision can give the same dyadic rearrangement. For instance, consider Figure 8, where both pairs represents the identity $\mathrm{id}\colon [0,1] \longrightarrow [0,1]$.



Figure 8

In general, *opposing carets* (symbol $\wedge$) have no effect on dyadic rearrangement. The latter correspond to an unnecessary "cut" from the subdivisions. See Figure 9. We say that two such pair of trees are *equivalent*, and that a pair of trees is *reduced* if it has no opposing carets. In other words, a reduced pair of trees is minimal in its equivalence class.



Figure 9

Proposition 2.5 (Theorem 1.2.4 in [Bel07])**.** *For each element of F one can associate a unique reduced pair of finite binary trees.*

In order to define Thompson's group T, we do the exact same construction, but starting with the unit circle $\mathbb{S}^1$ instead of the unit interval. Identifying $\mathbb{S}^1$ with $[0,1]/\{0,1\}$, the whole construction can be repeated almost readily.

More precisely, the definition of *dyadic subdivisions* of the circle is the same. The only small difference resides in the definition of *dyadic rearrangements* of the unit circle. In a dyadic rearrangement of the interval, the leftmost subinterval is forced to be mapped to the leftmost subinterval. On a circle, this is no longer true: a dyadic rearrangement of the circle has the extra freedom of *rotating* around the circle.

For example, the map

$$f\colon \mathbb{S}^1 \to \mathbb{S}^1\colon x \mapsto \lfloor x + \tfrac{1}{2} \rfloor$$

is a dyadic rearrangement from the subdivision $\mathbb{S}^1 = [0, \frac{1}{2}] \cup [\frac{1}{2}, 1]$ to itself.

In order to represent a dyadic rearrangement of the circle as a pair of opposing rooted binary trees, we thus need to remember to which subinterval the leftmost subinterval is mapped. We do so by adding a small bullet to the corresponding leaf of the target binary tree. For example, Figure 10 represents the above dyadic rearrangement.

Definition 2.6. Thompson's group T is the group of dyadic rearrangements of the circle.

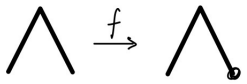


Figure 10

Clearly, dyadic rearrangements of the interval also give dyadic rearrangements of the circle after identifying $0 = 1$. This induces an inclusion of Thompson's groups $F \subset T$. In the *pairs of rooted tree picture*, elements of F correspond to elements of T with the added bullet on the leftmost leaf ot the target root.

Thompson's groups F and T are subgroups of an even larger group, Thompson's group V, which is obtained by giving even more freedom to *dyadic rearrangements*. Elements of F must fix the leftmost leaf (or subinterval), elements of T can move the leftmost leaf to another leaf, but need to preserve the cyclic order of the leaves. This naturally brings us to the definition of V, whose elements can now freely permute the leaves. Concretely, this means that elements of V are *not necessarily continuous* piecewise linear maps of $[0, 1]$ to itself, sending linearly a dyadic decomposition to another one. Allowing non-continuous piecewise linear maps gives the possibility of permuting the subintervals. In order to represent an element of V by a pair of opposing rooted binary trees, we now need to add the extra information of the corresponding permutation of the leaves.

Informally, we can think of the chain of Thompson's group $F \subset T \subset V$ as sitting over the chain $1 \subset \mathscr{C} \subset \mathscr{S}$, where $\mathscr{C}$ stands for cyclic group and $\mathscr{S}$ for symmetric group.

$$\begin{array}{ccccc} F & \hookrightarrow & T & \hookrightarrow & V \\ \vdots & & \vdots & & \vdots \\ 1 & \hookrightarrow & \mathscr{C} & \hookrightarrow & \mathscr{S} \end{array}$$

Of course, without more work the dotted lines cannot be turned into arrows since F, T, V contain dyadic rearrangements of dyadic subdivisions consisting of any number of subintervals.

The Houghton's groups were introduced by Houghton in [Hou78] and are popular for their finiteness properties and other characteristics. We give a brief definition of these groups that will appear in an example in the next section to motivate our study of asymptotically rigid mapping class group.

Let n be an integer strictly greater than two. Let X_n be $\{1, \cdots, n\} \times \mathbb{N}$. We denote by $\mathrm{Sym}(X_n)$ the group of all bijections on X_n, acting on the right on X_n. Let $(j, m) \in X_n$, one defines the elements $g_i \in \mathrm{Sym}(X_n)$ with $i = 2, \cdots, n$ by:

$$(j,m)g_i = \begin{cases} (1, m+1) & \text{if } j = 1, \\ (1,1) & \text{if } j = i, m = 1, \\ (i, m-1) & \text{if } j = i, m > 1, \\ (j,m) & \text{otherwise.} \end{cases}$$

Definition 2.7. The nth *Houghton group*, denoted H_n, is the subgroup of $\mathrm{Sym}(X_n)$ generated by $g_2, ..., g_n$.

In fact, an element $g \in \mathrm{Sym}(X_n)$ lies in H_n if and only if there exist $z_g^1, ..., z_g^n \in \mathbb{N}$ and $(t_g^1, ..., t_g^n) \in \mathbb{Z}^n$ such that, for all $i \in \mathbb{Z}$, $(i, m)g = (i, m + t_g^i)$ for all $m > z_g^i$. Hence t_g^i represents how far g moves the points $\{(i, m) \mid m > z_g^i\}$. Now observe that for each $i = 2, \cdots, n$,

$t^1_{g_i} = 1$ and $t^j_{g_i} = -\delta_{i,j}$ for $j = 2, \cdots, n$. There is a way to picture the g_i's. Let R_1 be the set of points $\{(1, m) \mid m \in \mathbb{N}\}$, we call it the first strand. We have n-strands counted clockwise. Then, Figure 11 from [Lee12] represents some elements of H_n.

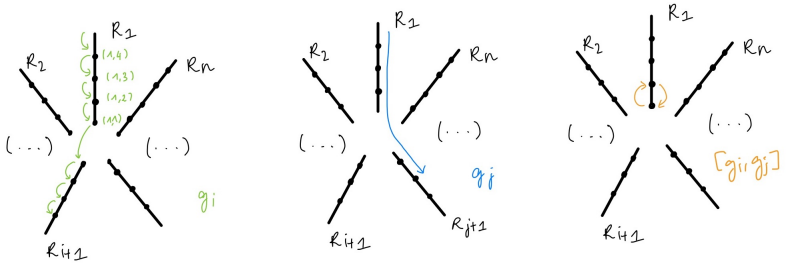


Figure 11

2.4 Cube complexes and median graphs

One of the key ideas from geometric group theory to understand a group is to study geometric objects on which the group acts. In particular, as geometric objects one can consider cube complexes. We give here a small toolbox that will be needed in the rest of the project.

Definition 2.8. A *cube complex* is a complex in which each cell is isometric to a Euclidean cube and the gluing maps are isometries.

Definition 2.9. A *median graph* is a graph in which every three vertices x, y, z have a unique median point, that is a vertex $m = m(x, y, z)$ lying on geodesic lines between each pair of x, y and z. Equivalently, m is the unique vertex satisfying

$$\begin{cases} d(x,y) = d(x,m) + d(m,y) \\ d(x,z) = d(x,m) + d(m,z) \\ d(y,z) = d(y,m) + d(m,z), \end{cases}$$

where d is the combinatorial distance. See Figure 12. Observe that a median graph is triangle-free.

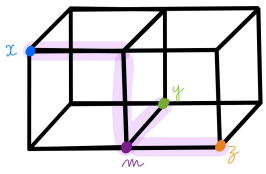


Figure 12

Definition 2.10. A *path* in a cube complex X is (a possibly infinite) sequence of vertices $x_0, x_1, ...$ such that two successive vertices of the sequence are adjacent in X (not identical). Let x and y be two vertices of X. The *distance* between x and y, denoted $d(x, y)$, is the minimal

length among the paths relating the two vertices. A path joining x and y with length $d(x, y)$ is called a *geodesic*. An infinite sequence of vertices $(x_i)_{i\in\mathbb{Z}}$ such that $d(x_n, x_m) = |n - m|$ is called an *infinite geodesic* (d is the combinatorial distance on the 0-skeleton).

Definition 2.11. Let X be a cube complex, a *morse function* on X is an affine function $f\colon X \longrightarrow \mathbb{R}$ non constant on each cube of positive dimension and such that the image of the 0-skeleton is discrete. Let $m \in \mathbb{R}$, the *sublevel set* X_m of X is the sub-complex generated by the vertices $y \in X$ satisfying $f(y) \leq m$.

Definition 2.12. Let X be a simplicial complex. The *link of a vertex* $x \in X$ is the simplicial complex denoted $\mathrm{Lk}(x, X)$ whose vertices correspond to edges in X that contains x. A k-simplex in $\mathrm{Lk}(x, X)$ is given by a collection of k edges in X containing x and spanning a k-cube. See Figure 13.

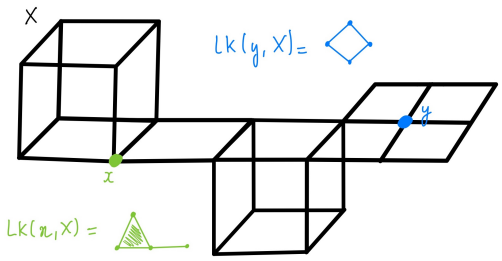


Figure 13

Definition 2.13. Let X be a cube complex and f a morse function on X. The *descending link of a vertex* $x \in X$ *with respect to* f is the link of x in $X_{f(x)}$.

Definition 2.14. Let X be a simplicial complex and $\Delta \subset X$ a simplex. The *link of the simplex* Δ is the subcomplex of X generated by all the simplices that are disjoint from Δ but spanning simplices with Δ.

The *star of the simplex* Δ is the union of all the simplices having Δ as a face.

Definition 2.15. Let Γ be a graph, its cube-completion $\Gamma^{\square}$ is the cube complex obtained from Γ by filling every sub-graph isomorphic to the 1-skeleton of a cube with a cube of the corresponding dimension.

Definition 2.16. A simplicial complex is *flag* if for every subset S of vertices such that every pair of vertices in S is in the complex, then the simplex with vertices S is also in the simplicial complex.

A cube complex is CAT(0) if it is simply connected and the link of every vertex is flag[1].

Proposition 2.17 (Theorem 6.1 in [Che00])**.** *Let X be a graph. Then $X^{\square}$ is a* CAT(0) *cube complex if and only if X is a median graph.*

We want to provide the proof of one direction that we will need in the next section. We show that if X is a median graph then $X^{\square}$ is a CAT(0) cube complex. In order to achieve this goal we start by giving some lemmata.

[1]This is not the usual definition, but it is equivalent.

Lemma 2.18. *Let X be a median graph. Then $X^{\square}$ is simply connected.*

Proof. Let C be a simple cycle of $X^{\square}$. By the skeletal approximation theorem we may assume that it is a cycle in X. We show that it can be deformed to an arbitrary vertex x in C. We proceed by induction on $\Sigma_{w\in C\cap X_0} d(x,w)$ where $d(\cdot,\cdot)$ is the combinatorial distance in X and X_0 is the 0-skeleton. Note that we do not have triangle since the graph is median. Take $y \in C$ at maximal distance from x and having neighbours $u, v \in C$ such that:

$$d(x,u) = d(x,v) < d(x,y).$$

There exists a such y since otherwise it means that we can find two points u and v at maximal distance that are adjacent. It contradicts the median condition on x, u, v. Since X is a median graph, there exists a unique point y' that is the median point between u, v, x, i.e. such that:

$$\begin{cases} d(u,v) = d(u,y') + d(y',v) \\ d(x,u) = d(x,y') + d(y',u) \\ d(x,v) = d(x,y') + d(y',v), \end{cases}$$

we observe that $d(x,y') = d(x,u) - \frac{1}{2}d(u,v) < d(x,y)$. Hence y' is closer to x than y. One has a square with vertices $y'uvy$ since $d(u,v) = 2$. This square gives us a homotopy from C to C' where the latter is the cycle obtained from C by replacing y with y'. We apply this procedure to all the points of the cycle at maximal distance from x. Then we can apply the induction hypothesis on C' to conclude. □

Lemma 2.19. *Let X be a median graph and Q a cube in $X^{\square}$. Let A and B be two vertices in Q, then any geodesic between A and B is inside Q.*

Proof. We proceed by induction on the dimension of Q. The one dimensional case is clear. Now assume that Q is n-dimensional. By contradiction we assume that there is a geodesic γ between some vertices A and B of Q which is not entirely contained in Q. If A and B share a face we are done by the induction hypothesis. Thus assume that $A = (0,...,0)$ and $B = (1,...,1)$.

- Assume that γ is not entirely outside Q and let P be a vertex on $\gamma \cap Q$ distinct from A and B. Then by induction the geodesics from A to P and from P to B must lie inside Q. Thus, γ is inside Q.

- Assume that γ is entirely outside Q and let D be the first vertex of the geodesic after A (so D is not in Q). Let C be the vertex labelled $(1,0,...,0) \in Q$ which is adjacent to A. See Figure 14 for a picture of the situation.

  Claim 2.20. The three vertices B, C, D have no median point.

  Proof. A geodesic λ from C to D has length less than or equal to two. We claim that λ has length two and must pass through A or through a vertex not in Q.

  – if λ has length one then it is an edge of X. So the latter contains a triangle ADC which is a contradiction since X is median.

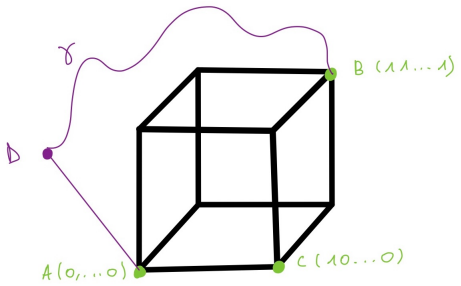


Figure 14

– Assume that λ has length two and E is the vertex of λ between C and D lying in Q and is distinct from A.
  * If $d(A, E) = 1$, then X contains a triangle AED.
  * If $d(A, E) = 2$, then X contains a copy of $K_{2,3}$ given by the square in Q induced by the three vertices ACE and the path ADE.
  * if $d(A, E) \geq 3$, one obtains a contradiction because then $d(E, C) \geq 3$.

Consequently, λ has length two and must pass through A or through a vertex not in Q. Then, by induction, any geodesic from B to C lies entirely inside Q. Thus the only candidate for a median vertex for B, D, C is C. If C is the median vertex then one has a geodesic from D to B passing through C. On one hand we know that γ has length less than or equal to n so a geodesic from D to B has length less than or equal to $n-1$. On the other hand, a geodesic from D to C has length 2 and a geodesic between C and B has length $n-1$. Thus one has a contradiction and C is not a median vertex. Hence B, D, C have no median vertex. □

By the previous claim we get a contradiction and γ cannot be entirely outside Q.

Consequently, γ lies inside Q. □

Definition 2.21. Let X be a cube complex. A 3-*corner* in X, consists of 3 pairwise distinct squares sharing a vertex and pairwise sharing an edge and not contained in a 3-cube of X. The common vertex is called the *root of the* 3-*corner*.

A 1-*corner* is a square with two adjacent edges identified (i.e. a cone without the base).

A 2-*corner* consists of two squares sharing two consecutive edge. Having a 2-*corner* implies that the 1-skeleton contains a copy of the bipartite graph $K_{3,2}$.

Lemma 2.22. *Let X be a median graph. Then there is no* 3-*corner in $X^{\square}$.*

Proof. Let C be a 3-corner in $X^{\square}$. Label the vertices of C by bit strings $e = (e_1, e_2, e_3)$ with $e_i \in \{0, 1\}$, so that two adjacent vertices differ by one bit, as in Figure 15. Let A be the median vertex between 110, 101 and 011.

Claim 2.23. A is distinct from all the other vertices in C.

Proof. To see this, first observe that the distance between two vertices among 110, 101 and 011 is always two. Indeed, we note that the distance is less than or equal to two. Then, if two

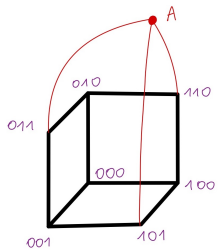


Figure 15

of the previous vertices are at distance one then there is a triangle which is a contradiction because X is median.

Then, assume by contradiction that A lies in C. By definition A lies on the geodesics linking pairwise 110, 101 and 011. It cannot be 000, since that vertex is on none of those geodesics. Thus it is either 011, 110, 101, 010, 100, or 001. For example, if A is 011 there is a contradiction since no geodesic between 101 and 110 passing through 011. Similarly if A is 010 there is a contradiction since no geodesic between 100 and 001 passing through 010. Idem for the other cases. So A is not in C. □

Consequently A lies outside C and is at distance one from 110, 101 and 011. Hence our 3-corner C is in fact contained in a 3-cube, thus it is not a 3-corner. □

Lemma 2.24. *Let X be a median graph. If three 3-cubes Q_1, Q_2 and Q_3 of $X^{\square}$ share a common 1-cube Q and pairwise share common 2-cubes $Q_{i,j}$ where $i, j = 1, 2, 3$ then they are contained in a 4-cube.*

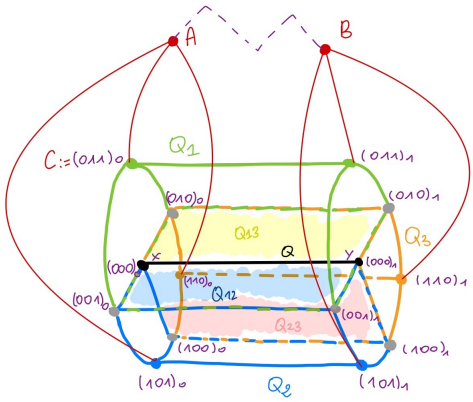


Figure 16

Proof. Let x and y be the edge Q. Consider the situation and the notation in Figure 16. Observe that we label the vertices of the two 3-corner (C_0 and C_1) by bit strings $v = (e_1, e_2, e_3)_k$ with $e_i \in \{0, 1\}$, $k = 0, 1$ in a particular way and so that two adjacent vertices differ by one bit. Let A be the unique median vertex between the three vertices $(011)_0$, $(110)_0$ and $(101)_0$ from C_0 and B the one between $(011)_1$, $(110)_1$ and $(101)_1$ from C_1.

First, observe that A (resp. B) does not belong to C_1 (resp. C_0) because otherwise one has a copy of $K_{2,3}$. For instance, given by $A = (011)_1, (010)_0, (110)_0, (100)_0, (000)_0$. Since

X is median this is a contraction. Indeed, then three vertices in the copy of $K_{2,3}$ with two incident edges do not have a unique median vertex.

Secondly, A is distinct from B since otherwise we have a triangle in X, for instance given by $(011)_0$, $(011)_1$ and $A = B$. This is a contradiction because X is median.

Finally, A and B are connected by an edge. Indeed, we know that $d(A, B) \leq 3$ since $A - (011)_0 - (011)_1 - B$ is a path in X. If $d(A, B) = 1$ we are done. If $d(A, B) = 2$ then the median between A, B and $C := (011)_0$ does not exists. Indeed, the only two geodesics between C and D do not intersect the two geodesics from B to C and from B to D. If $d(A, B) = 3$ we obtain a similar contradiction. □

Partial proof of Proposition 2.17. We show one direction of the proposition. Let X be a median graph, we want to show that $X^{\square}$ is a CAT(0) cube complex. By Lemma 2.18 the latter is simply connected. It remains to show that the link of each vertex is flag, which is equivalent (Proposition 3.3 in [Che00]) to showing that if three $(n+2)$-cubes Q_1, Q_2 and Q_3 of $X^{\square}$ share a common n-cube Q and pairwise share common $(n+1)$-cubes $Q_{i,j}$ then they are contained in a $(n+3)$-cube. The case $n = 0$ is the content of Lemma 2.22 while the case $n = 1$ is the one of Lemma 2.24. Now we consider the general case. See Figure 17. Note that we use

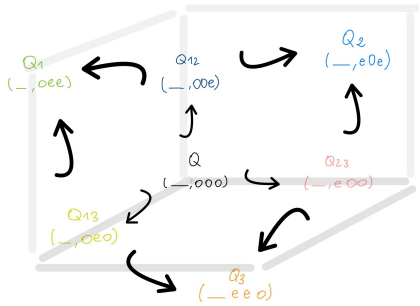


Figure 17

a particular notation, we label coherently the vertices of the Q_i by bit strings $(e_1, ..., e_{n+3})$ with $e_i \in \{0, 1\}$, so that two adjacent vertices differ by one bit. Let $(e_1, ..., e_n) \in \{0, 1\}^n$ then by the case $n = 0$ the three vertices $(e_1, ..., e_n, 1, 1, 0)$, $(e_1, ..., e_n, 1, 0, 1)$, and $(e_1, ..., e_n, 0, 1, 1)$ have a unique median vertex completing the 3-corner, we call it $(e_1, ..., e_n, 1, 1, 1)$.

Claim 2.25. The new vertices $(e_1, ..., e_n, 1, 1, 1)$ are all distinct from each other.

Proof. We proceed by induction on n. We already treated the base case. By contradiction, let us pick two distinct elements $\underline{e}, \underline{e}'$ from $\{0, 1\}^n$ such that $(\underline{e}, 1, 1, 1) = (\underline{e}', 1, 1, 1)$. If $\underline{e}$ and $\underline{e}'$ share a common digit then they are contained in a $(n-1)$-cube and we are done by the induction hypothesis. Otherwise, we assume that $\underline{e} = (0, ..., 0)$ and $\underline{e}' = (1, ..., 1)$. Recall that $(0, ..., 0, 1, 1, 1) = (\underline{e}, 1, 1, 1) = (\underline{e}', 1, 1, 1) = (1, ..., 1)$. Thus, $(0, ..., 0, 1, 1, 0)$ and $(1, ..., 1, 1, 1, 0)$ are at distance less than or equal to two in X. But the previous vertices are in Q_3 so by Lemma 2.19 they are at distance less than or equal to two in Q_3. Thus $\underline{e}$ and $\underline{e}'$ differ by at most two digits so $n \leq 2$. If $n = 1$ this is the base case. If $n = 2$, we have a path $(0, ..., 0, 1, 1, 0) - (0, ..., 0, 1, 1, 1) - (1, ..., 1, 1, 1, 1) - (1, ..., 1, 1, 1, 0)$. The latter is a geodesic between $(0, ..., 0, 1, 1, 0)$ and $(1, ..., 1, 1, 1, 0)$ not contained in the red cube in Figure

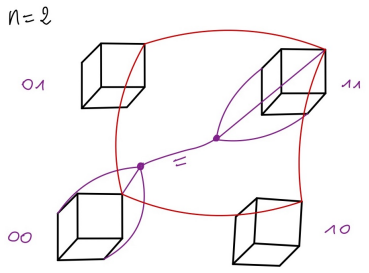


Figure 18

18, whereas its endpoints lie in the red cube. Thus we have a contradiction with Lemma 2.19. □

Claim 2.26. The new vertices $(e_1, ..., e_n, 1, 1, 1)$ are not already in $Q_1 \cup Q_2 \cup Q_3$.

Proof of Claim 2.26. We want to show that the vertices $(e_1, ..., e_n, 1, 1, 1)$ are not in $Q_1 \cup Q_2 \cup Q_3$. For simplicity we consider $(0, ..., 0, 1, 1, 1)$ and assume by contradiction that it belongs to $Q_1 \cup Q_2 \cup Q_3$. Say $(0, ..., 0, 1, 1, 1) = (f_1, .., f_n, f_{n+1}, f_{n+2}, f_{n+3})$ with $(f_{n+1}, f_{n+2}, f_{n+3}) \neq (1, 1, 1)$.

Without loss of generality, take $f_{n+3} \neq 1$, then $(0, ..., 0, 1, 1, 1) = (f_1, .., f_n, f_{n+1}, f_{n+2}, 0)$. Lemma 2.19, gives an edge S from $(0, ..., 0, 1, 1, 0)$ to $(0, ..., 0, 1, 1, 1) = (f_1, .., f_n, f_{n+1}, f_{n+2}, 0)$ which is in Q_3 because the end vertices are in Q_3. Thus, $(0, ..., 0, 1, 1)$ and $(f_1, ..., f_n, f_{n+1}, f_{n+2})$ differ by one digit. Consequently,

- Either $(f_1, .., f_n, f_{n+1}, f_{n+2}) = (0, ..., 0, 1, 0)$. In this case, $(0, ..., 0, 1, 1, 1) = (f_1, .., f_{n+3}) = (0, ..., 0, 1, 0, 0)$. Thus the edge S joins two vertices of the 3-cube $(0, ..., 0, e, e, e)$ and we are done by the base case $n = 0$. See the claim.

- Or $(f_1, .., f_n, f_{n+1}, f_{n+2}) = (1, 0, ..., 0, 1, 1)$. In this case, $(0, ..., 0, 1, 1, 1) = (f_1, .., f_{n+3}) = (1, 0, ..., 0, 1, 1, 0)$. Thus the edge S joins two vertices of the 3-cubes $(0, ..., 0, e, e, e)$ and $(1, 0, ..., 0, e, e, e)$ and we are done by the base case $n = 1$.

□

Now, we prove one implication in Proposition 2.17. Because of the two claims above, it only remains to show that all the edges from $(e_1, ..., e_i = 0, ..., e_{n+3})$ to $(e_1, ..., e_i = 1, ..., e_{n+3})$ are in X.

- if $i \leq n$ and $(..., e_{n+1}, e_{n+2}, e_{n+3}) \neq (...1, 1, 1)$, the edges are already in $Q_1 \cup Q_2 \cup Q_3$ by construction.

- if $i > n$ (say $i = n + 1$) and $(..., e_{n+2}, e_{n+3}) \neq (...1, 1)$, the edges are already in Q_2 or Q_3 since the edges from $(e_1, ..., e_n, 0, e_{n+2}, e_{n+3})$ to $(e_1, ..., e_n, 1, e_{n+2}, e_{n+3})$ are in Q_2 or Q_3 by construction.

- if $i > n$ (say $i = n + 1$) and $(..., e_{n+2}, e_{n+3}) = (...1, 1)$, the edge from $(e_1, ..., e_n, 0, 1, 1)$ to $(e_1, ..., e_n, 1, 1, 1)$ is in X by case $n = 0$ and the construction of $(e_1, ..., e_n, 1, 1, 1)$.

- if $i \leq n$ and $(...e_{n+2}, e_{n+3}) = (...1, 1, 1)$, for instance if $i = n$ (other cases are similar), then we are looking for the edge from $(e_1, ..., e_{n-1}, 0, 1, 1, 1)$ to $(e_1, ..., e_{n-1}, 1, 1, 1, 1)$ which is in X by the case $n = 1$

□

Proposition 2.27 (Theorem 3.3.1 in [Gen22])**.** *A graph is median if and only if its cube-completion is simply connected and has no corner.*

The next Proposition provides a mean to show that a cube complex is CAT(0).

Proposition 2.28. *Let X be a connected graph. Assume that:*

1. *the cube completion $X^{\square}$ is simply connected,*
2. *$X^{\square}$ has no 1-corner.*
3. *$X^{\square}$ also satisfies the 3-cube condition (every 3-corner must span a 3-cube),*
4. *X does not contain a copy of the complete bipartite graph $K_{3,2}$.*

Then, $X^{\square}$ is a CAT(0) *cube complex.*

Proof. Let X be a connected graph satisfying the assumptions in the statement above. First, we observe that by the assumptions (1), $X^{\square}$ is simply connected and since it does not contain a copy of $K_{3,2}$ (4) it does not have 2-corner. Moreover, by assumptions (2) and (3), $X^{\square}$ does not have any 1-corner and 3-corner. By Proposition 2.27 we conclude that X is a median graph. Then, by Proposition 2.17, we conclude that $X^{\square}$ is CAT(0). □

Definition 2.29. The automorphism group of a cube complex X is the set of isometries X to itself.

Definition 2.30. Let X be CAT(0) cube complex and G a group acting on it. Then every element in G gives an automorphism on X, it is called *elliptic* if it has a fixed point in the 0-skeleton of X.

Definition 2.31. Let X be CAT(0) cube complex and G a group acting on it. An non-elliptic automorphism $f \in G$ of X is *loxodromic* if it preserves an infinite geodesic γ and if it acts on γ as a non trivial translation.

Definition 2.32. Let X be CAT(0) cube complex and p be a vertex in X and f be a loxodromic element in $\mathrm{Aut}(X)$. We denote by $\delta(f)$ the minimal distance between a vertex of X and its image under f, it is called the *translation length* of f.

3 Studying the CAT(0)ness of the cube complex $\mathscr{C}(A_{n,m})$

To obtain nice properties about $\mathfrak{mod}(A_{n,m})$ we want to find a nice cube complex on which $\mathfrak{mod}(A_{n,m})$ acts. More precisely, we want to find a CAT(0) cube complex. In [GLU22], the authors give a cube complex $\mathscr{C}(A_{n,m})$ associated to $\mathfrak{mod}(A_{n,m})$. However, in Section 3.3, the authors also show that $\mathscr{C}(A_{1,3})$ is not CAT(0). In the first and second parts of this section, we derive some useful properties in order to study the CAT(0)ness of the latter cube complex.

3.1 Definitions and properties

Let A be a locally finite planar tree. We define a relation on the set of pairs (Σ, ϕ), where Σ is an admissible surface of $\mathscr{S}^\sharp(A)$ and ϕ lies in $\mathfrak{mod}(A)$. Two pairs are related $(\Sigma, \phi) \sim (\Sigma', \phi')$ if and only if $\phi'^{-1}\phi$ is isotopic to an asymptotically homeomorphism rigid outside Σ and mapping Σ to Σ'. Observe that $(\Sigma, \phi) \sim (\Sigma, \phi')$ automatically if ϕ is isotopic to ϕ', since then $\phi'^{-1}\phi$ is isotopic to the identity.

It is clear that the relation $\sim$ is an equivalence relation.

Definition 3.1. Let $\mathscr{C}(A)$ be the cube complex given by:

- vertices: the equivalence classes $[\Sigma, \phi]$ with $\Sigma \subseteq \mathscr{S}^\sharp(A)$ an admissible surface and ϕ in $\mathfrak{mod}(A)$.
- edges: there is an edge between $[\Sigma, \phi]$ and $[\Sigma \cup H, \phi]$ where H is an adjacent polygon to Σ.
- cubes: For any vertex $[\Sigma, \phi]$, and distinct adjacent polygons $H_1, ..., H_k$ to Σ, the subgraph $\left\{[\Sigma \cup \bigcup_{i\in I} H_i, \phi] | I \subset \{1, ..., k\}\right\}$ is filled with a k-cube.

Let $g \in \mathfrak{mod}(A)$ and $[\Sigma, \phi]$ a vertex in $\mathscr{C}(A)$, we define the action of g on $[\Sigma, \phi]$ by:

$$g \cdot [\Sigma, \phi] := [\Sigma, g \circ \phi].$$

Proposition 3.2. *Let A be a locally finite planar tree. Then $\mathfrak{mod}(A)$ acts on $\mathscr{C}(A)$.*

Proof. The action defined above is well-defined. Take g in $\mathfrak{mod}(A)$ and let $[\Sigma, \phi], [\Sigma', \phi']$ be two representatives of the same equivalence class. Then, $g \cdot [\Sigma, \phi] = g \cdot [\Sigma', \phi']$. Indeed, $(g\phi')^{-1} g\phi = \phi'^{-1}\phi$, which is isotopic to an asymptotically rigid homeomorphism sending Σ to Σ' and which is rigid outside Σ. □

Lemma 3.3. *Let A be a locally finite tree, $x = [\Sigma, \phi]$ and y two vertices of $\mathscr{C}(A)$ which are adjacent. If $h(y) > h(x)$, then there exists H an adjacent polygon to Σ such that $y = [\Sigma \cup H, \phi]$.*

Proof. Since x and y are adjacent, by definition, there exist an admissible surface Δ, I an adjacent polygon to Δ and ψ in $\mathfrak{mod}(A)$ such that $x = [\Delta, \psi]$ and $y = [\Delta \cup I, \psi]$. From $[\Sigma, \phi] = [\Delta, \psi]$ we deduce that $\phi^{-1}\psi$ is isotopic to an asymptotically rigid homeomorphism mapping Δ to Σ and rigid outside Δ. Hence, I is sent by ϕ to another polygon of the rigid structure called H which is adjacent to Σ because of the rigidity. Thus $y = [\Sigma \cup H, \phi] = [\Delta \cup I, \psi]$. □

In Section 3.3 of [GLU22] it is shown that the cube complex $\mathscr{C}(A_{n,m})$ may not be CAT(0). We want to know for which family of trees our cube complex is CAT(0).

Let $n, m \geq 1$ and consider the cube complex $\mathscr{C}(A_{n,m})$. *The orientation of $\mathscr{C}(A_{n,m})$ with respect to the height of vertices*, or *height-orientation*, is obtained by giving an orientation to all the edges of $\mathscr{C}(A_{n,m})$ from lower height to higher height. That is, take an edge $e = (x, y)$, then e is oriented with an arrow from x to y if and only if $h(x) > h(y)$, otherwise it is oriented from y to x. Given a vertex x, we say that an incident edge of x *goes out* if it is oriented from x to another vertex, otherwise it *goes in*. Figure 19 represents a height-oriented edge.

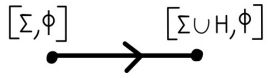


Figure 19

Lemma 3.4. *Let $n, m \geq 1$ and let C be a 2-cube in $\mathscr{C}(A_{n,m})$. Then, C has a unique height-orientation. More precisely, one of the vertices has some height $h \in \mathbb{N}^*$, its two adjacent vertices have height $h + 1$ and the remaining vertex has height $h + 2$.*

Proof. Orient each edge of C with respect to the height, this brings us to distinguish four cases.

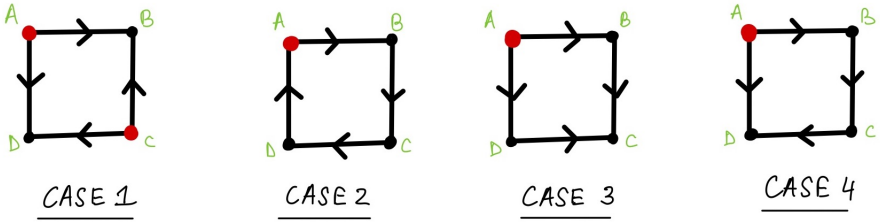


Figure 20

Consider Figure 20. Applying Definition 3.1 and Lemma 3.3 (starting from the red points) allows us to keep only the third case which is a square of the desired form.

- Indeed, consider the first case, we label the vertex A (resp. C) as $[\Sigma, \mathrm{id}]$ (resp. $[\Gamma, \pi]$). On one hand, by Lemma 3.3 applied from A and C at vertex B one has, $[\Sigma \cup H_1, \mathrm{id}] = [\Gamma \cup I_1, \pi]$. That is, $\phi(\Gamma \cup I_1) = \Sigma \cup H_1$ and ϕ is rigid outside $\Gamma \cup I_1$. On the other hand, by Lemma 3.3 applied from A and C at vertex D one has, $[\Sigma \cup H_2, \mathrm{id}] = [\Gamma \cup I_2, \pi]$. By definition, $\phi(\Gamma \cup I_2) = \Sigma \cup H_2$ and ϕ is rigid outside $\Gamma \cup I_2$. Thus $\phi(H_i) = I_i$ for $i = 1, 2$. Consequently, $[\Gamma, \phi] = [\Sigma, \mathrm{id}]$, which is a contradiction.

- For the second case, denote $[\Sigma, \mathrm{id}]$ the red vertex and apply Lemma 3.3 successively from low height to high height. Then we obtain $[\Sigma, \mathrm{id}] = [\Sigma \cup H_1 \cup H_2 \cup H_3 \cup H_4, \mathrm{id}]$, which is a contradiction.

- One can exclude the fourth case by proceeding similarly to the third case.

□

The cube complex $\mathscr{C}(A)$ is the cube completion of its underlying graph. This is the content of the following proposition.

Proposition 3.5. *Let A be a locally finite planar tree. Let Γ_A be the underlying graph of $\mathscr{C}(A)$. Then, $\Gamma_A^{\square} = \mathscr{C}(A)$.*

Proof. The inclusion $\mathscr{C}(A) \subset \Gamma_A^{\square}$ follows from the definitions. Let C be an n-cube in $\Gamma_A^{\square}$. We show by induction that its vertices form an n-cube in $\mathscr{C}(A)$ as well. In other words, we need to show that its vertices have the form $\{[\Sigma \cup \bigcup_{i\in I} H_i, \phi] \quad | \quad I \subset \{1, \dots, n\}\}$ for some admissible surface Σ, an asymptotically rigid homeomorphism ϕ, and distinct polygons $H_1, \dots, H_n$ adjacent to Σ.

For $n = 1$, this is the content of Lemma 3.3. We will need the base case $n = 2$ in the proof, treated by Lemma 3.4.

Claim 3.6. If the vertices of C are labelled by bit strings $\underline{e} = e_1 \dots e_n$ with $e_i \in \{0, 1\}$ so that the height function h atteins a mimimum at $\underline{0} = 0 \dots 0$, (and two adjacent vertices differ by one bit) then

$$h(\underline{e}) = h(\underline{0}) + \textstyle\sum_{i=1}^{n} e_i$$

for each $\underline{e}$. In particular, the function h has a unique global minimum and a unique global maximum on the vertices of C.

Observe that, along any edge, the height must increase by 1 or decrease by 1.

Proof of the claim. We show this by induction on n. For $n = 1$ and $n = 2$, this directly follows from the above. Assume that it holds for any $(n-1)$-cube and consider the n-cube C.

Pick a vertex in C with minimal height and give it the label $\underline{0} = 0 \dots 0$. Label the rest of the cube coherently. Any vertex of C except $\underline{1} = 1 \dots 1$ shares an $(n-1)$-cube with $\underline{0}$, hence by induction hypothesis we have

$$h(\underline{e}) = h(\underline{0}) + \textstyle\sum_{i=1}^{n} e_i$$

for all $\underline{e} \neq \underline{1}$. In particular, all the neighbours of $\underline{1}$ have height $h(\underline{0}) + (n-1)$. Thus $\underline{1}$ must have height $h(\underline{0}) + n$ or $h(\underline{0}) + (n-2)$. The latter cannot happen. Indeed it would mean that one can find a 2-cube C' containing $\underline{1}$, two of its neighbours and a neighbour of them so that we would have two points with maximal height in C', as in Case 1 of Figure 20 [for example, the 2-cube with vertices $001 \dots 1$, $101 \dots 1$, $011 \dots 1$ and $111 \dots 1$]. Thus $\underline{1}$ has height $h(\underline{0}) + n$ and we are done. □

Fix a labelling of the vertices of C as in the above claim. Then, by 3.3, the n neighbours of $\underline{0}$, i.e. the vertices with exactly one 1 in their label, must correspond to the attachment of n distinct polygons to Σ. More precisely, the vertex with the 1 in the i-th entry has the form $[\Sigma \cup H_i, \phi]$, and the polygons $H_1, \dots, H_n$ are all distinct and adjacent to Σ.

Now, the vertices with exactly two 1's in their label are also determined by the above. For example, the vertex with 1's in entries i and j must have the form $[\Sigma \cup H_i \cup H_j, \phi]$. This follows from the base case $n = 2$.

Following this argument for the labels with exactly three 1's, then four, and so on, we are able to identify all vertices in the cube C (each time using the base case $n = 2$). More precisely, the vertex whose label has 1's in entries $i_1, \dots, i_k$ must have the form $[\Sigma \cup H_{i_1} \cup \dots \cup H_{i_k}, \phi]$. This concludes the proof. □

Proposition 3.5 will allow us to use Proposition 2.28 with $\mathscr{C}(A_{n,m})$ instead of $A'^{\square}_{n,m}$ in order to study the CAT(0)ness of the cube complex $\mathscr{C}(A_{n,m})$. In the rest of this subsection, we give some properties of $\mathscr{C}(A_{n,m})$ that will help us to achieve our goal.

Lemma 3.7. *For all $n, m \geq 1$, there is no 1-corner in the 1-skeleton of $\mathscr{C}(A_{n,m})$.*

Proof. Recall that a 1-corner is a square with two adjacent edges identified. Thus if one has a 1-square then one has two edges with their two endpoints identified. This is a contradiction with the height-orientation since one has one of the two cycle in Figure 21. □

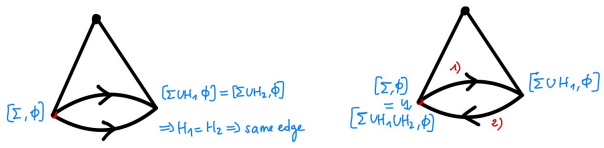


Figure 21

Lemma 3.8. *For all $n, m \geq 1$, there is no copy of the bipartite graph $K_{2,3}$ in the 1-skeleton of $\mathscr{C}(A_{n,m})$.*

Proof. Assume by contradiction that such a copy exists. Again we give the height-orientation to each edge of the copy. This leads us to distinguish eight cases. We show that each case gives a contradiction. Consider Figure 22. By Lemma 3.4, all the cases, except Case 4, give

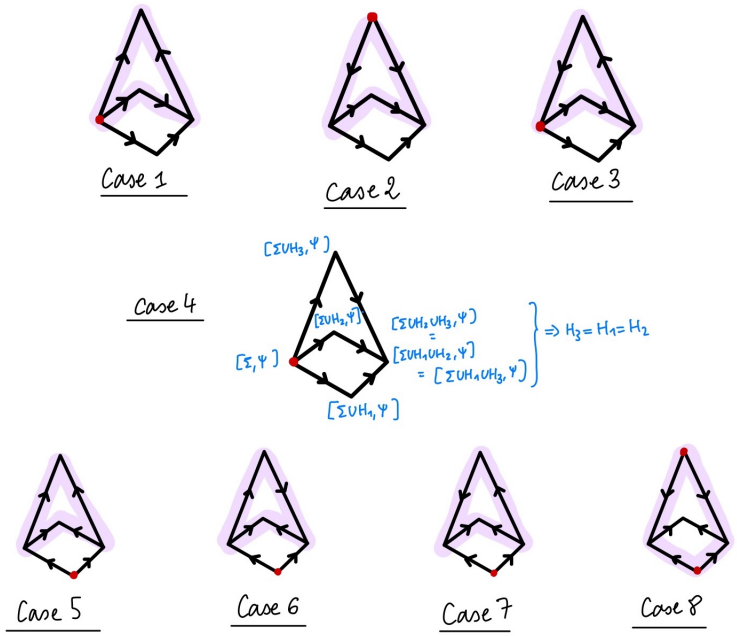


Figure 22

a contradiction. Indeed, in these situations one has a square of the same form as Case 1, 2 or 3 from Lemma 3.4 and we already eliminated these cases. Now, it remains to study Case 4. One marks by a red point the vertex having all its incident edges going out. Then,

we label this red point $[\Sigma, \psi]$. By Lemma 3.3, we label its adjacent vertices as $[\Sigma \cup H_1, \psi]$, $[\Sigma \cup H_2, \psi]$ and $[\Sigma \cup H_3, \psi]$. Finally, the remaining vertex with all incident edges going inside is $[\Sigma \cup H_1 \cup H_2, \psi] = [\Sigma \cup H_2 \cup H_3, \psi] = [\Sigma \cup H_1 \cup H_3, \psi]$. Hence $H_1 = H_2 = H_3$, which is a contradiction since the vertices are all distinct. □

Definition 3.9. Let $\mathscr{C}(A_{n,m})$ be height-oriented. A vertex x in $\mathscr{C}(A_{n,m})$ has *type* (a, b) if x has a incident edges going out and b incident edges going inside. We say that a 3-*corner is of type* (a, b) if its root is a vertex of type (a, b).

Lemma 3.10. *For all $n, m \geq 1$, the 3-corners of type $(3, 0)$, $(2, 1)$ or $(1, 2)$ in $\mathscr{C}(A_{n,m})$ can be completed into a 3-cube.*

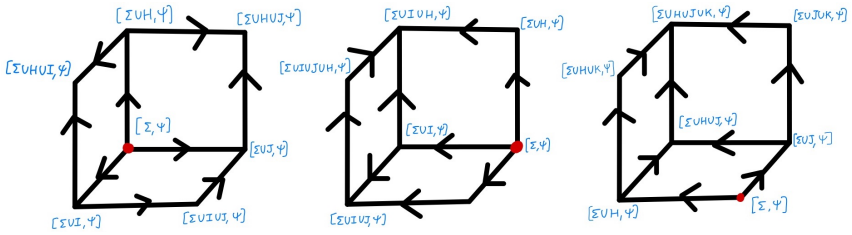


Figure 23

Proof. Let $n, m \geq 1$, in Figure 23 we represent three 3-corners of type $(3, 0)$, $(2, 1)$ and $(1, 2)$ in $\mathscr{C}(A_{n,m})$. For each case we need to find a vertex to complete the 3-corner into a 3-cube. One marks by a red point the vertices having all their incident edges going out.

- Consider Case 1 in Figure 23. Here one has only one red point. We label the latter $[\Sigma, \psi]$. By Lemma 3.3, the adjacent vertices can be labelled as $[\Sigma \cup H, \psi]$, $[\Sigma \cup J, \psi]$ and $[\Sigma \cup I, \psi]$. Then, again by Lemma 3.3, the common adjacent vertices of the previous vertices can be labelled as $[\Sigma \cup H \cup J, \psi]$, $[\Sigma \cup H \cup I, \psi]$ and $[\Sigma \cup I \cup J, \psi]$. Now, we take the vertex $[\Sigma \cup H \cup I \cup J, \psi]$ to complete the 3-corner into a 3-cube.

- Now we consider the situation described in Case 2 in Figure 23. There is only one red point and we label it as $[\Sigma, \psi]$. Then, by Lemma 3.3, its adjacent vertices can be labelled as $[\Sigma \cup H, \psi]$, $[\Sigma \cup J, \psi]$ and $[\Sigma \cup I, \psi]$. As for the case above, the common adjacent vertices of the previous vertices can be labelled as $[\Sigma \cup H \cup J, \psi]$, $[\Sigma \cup H \cup I, \psi]$ and $[\Sigma \cup I \cup J, \psi]$. It remains to take the vertex $[\Sigma \cup H \cup J, \psi]$ to complete the 3-corner into a 3-cube.

- For the third case in Figure 23 we proceed in the same way as above. We label the red point as $[\Sigma, \psi]$. Again by Lemma 3.3, the adjacent vertices of the latter are denoted by $[\Sigma \cup H, \psi]$ and $[\Sigma \cup J, \psi]$. Consequently the root of the 3-corner is $[\Sigma \cup H \cup J, \psi]$. We label by $[\Sigma \cup H \cup K_1, \psi]$ resp. $[\Sigma \cup J \cup K_2, \psi]$ the remaining adjacent vertices of $[\Sigma \cup H, \psi]$ resp. $[\Sigma \cup J, \psi]$. Now, the common adjacent vertices of $[\Sigma \cup H \cup K_1, \psi]$, $[\Sigma \cup J \cup K_2, \psi]$ and $[\Sigma \cup H \cup J, \psi]$ must have an height equals to $h(\Sigma) + 3$ thus $K_1 = K_2$. Hence the vertex where all the edges are going inside is $[\Sigma \cup H \cup J \cup K, \psi]$. Now, we take $[\Sigma \cup K, \phi]$ to complete the 3-corner into a 3-cube.

□

3.2 Completing an attracting root

Recall that we study the CAT(0) of $\mathscr{C}(A_{n,m})$ with Proposition 2.28. In particular we want that all the 3-corners can be completed into 3-cubes. In this section, we denote by M the central polygon in $\mathscr{C}(A_{n,m})$, which is the polygon containing the puncture corresponding to the unique vertex of valence m. If $m = n + 1$, the central polygon is an arbitrary polygon that we fix.

Definition 3.11. A root of type $(0, 3)$ in a 3-corner of $\mathscr{C}(A_{n,m})$ is called an *attracting root*.

The goal of this subsection is to show the following proposition:

Proposition 3.12. *Let $n > 1$ and $1 \leq m \leq n + 1$. Then, one can complete the 3-corner associated to an attracting root in $\mathscr{C}(A_{n,m})$ into a 3-cube.*

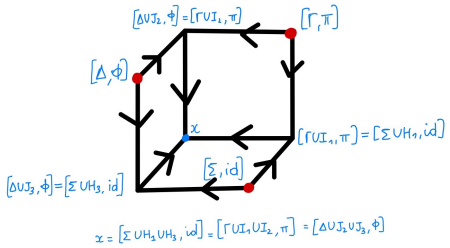


Figure 24

First, we consider the general situation of a 3-corner having an attracting root in Figure 24, where we applied Lemma 3.3 successively starting from the three red points. Then, we draw up the inventory of the facts that follow from the equalities associated to each vertex in Figure 24.

(i) $\pi(\Gamma \cup I_1) = \Sigma \cup H_1$ and π is rigid outside $\Gamma \cup I_1$,

(ii) $\phi(\Delta \cup J_3) = \Sigma \cup H_3$ and ϕ is rigid outside $\Delta \cup J_3$,

(iii) $\pi^{-1}\phi(\Delta \cup J_2) = \Gamma \cup I_2$ and $\pi^{-1}\phi$ is rigid outside $\Delta \cup J_2$,

(iv) $\pi(\Gamma \cup I_1 \cup I_2) = \Sigma \cup H_1 \cup H_3$ and π is rigid outside $\Gamma \cup I_1 \cup I_2$,

(v) $\phi(\Delta \cup J_2 \cup J_3) = \Sigma \cup H_1 \cup H_3$ and ϕ is rigid outside $\Delta \cup J_2 \cup J_3$,

(vi) $\pi^{-1}\phi(\Delta \cup J_2 \cup J_3) = \Gamma \cup I_1 \cup I_2$ and $\pi^{-1}\phi$ is rigid outside $\Delta \cup J_2 \cup J_3$.

Now, we distinguish five configurations of 3-corners whose roots are of type $(0, 3)$ in $\mathscr{C}(A_{n,m})$.

Definition 3.13. Consider the general situation above of a 3-corner with an attracting root. Then, one says that the 3-corner is of

- *Type A* if Δ, Σ and Γ do not contain the central polygon and the H's, I's and $J's$ are not M.

- *Type B* if Δ, Σ and Γ do not contain the central polygon and among the H's, I' and $J's$ some are M.
- *Type C* if Δ, Σ and Γ contain the central polygon.
- *Type D* if Σ contains M but Δ and Γ do not.
- *Type E* if Σ does not contain M but Δ and Γ do.

In order to prove Proposition 3.12 we need the next three propositions:

Proposition 3.14. *Consider a 3-corner in $\mathscr{C}(A_{n,m})$ and assume that we are in one of the following situations.*

- *Let $n > 1$, $m \geq 1$ and the 3-corner has an attracting root of Type A, B, or D.*
- *Let $n > 1$ and $1 \leq m \leq n+1$ and the 3-corner has an attracting root of Type C.*

Then, our 3-corner can be completed into a 3-cube.

Proposition 3.15.
- *Let $n > 1$, $m \geq 1$ and $m \neq n+1$. Then, there is no 3-corner with an attracting root of Type E in $\mathscr{C}(A_{n,m})$*
- *Let $n > 1$ and $m = n+1$. Then, every 3-corner in $\mathscr{C}(A_{n,m})$ with an attracting root and of Type E can be completed into a 3-cube.*

Remark 3.16. The situation in the second part of Proposition 3.15 is exactly the same as situation in Proposition 3.14 in the particular case of Type A. Indeed, if $m = n+1$ then all the polygons of the rigid structure are the same so we do not distinguish a central one.

We need the following lemmata to prove Proposition 3.14.

Lemma 3.17. *Let $n > 1, m \geq 1$ and assume that we are in the situation of Figure 24. The image of the polygon I_1 by π is contained in Σ and shares $\sharp\,\mathrm{Fr}(I_1) - 1$ frontier-arcs with polygons outside Σ and distinct from H_1 and H_3.*

Proof. Facts (i) and (iv) imply that $\pi(I_2) = H_3$. Similarly (ii) and (v) (resp. (iii) and (vi)) give that $\phi(J_2) = H_1$ (resp. $\pi^{-1}\phi(J_3) = I_1$).

Thus one has the following diagram:

$$\begin{array}{ccccc} J_3 & \xrightarrow{\pi^{-1}\phi} & I_1 & \xrightarrow{\pi} & \pi(I_1) \subseteq_{(i)} \Sigma \cup H_1 \\ & \searrow^{\phi} & & & \| \\ & & & & \phi(J_3) \subseteq_{(ii)} \Sigma \cup H_3 \end{array}$$

we deduce from this diagram that $\pi(I_1) = \phi(J_3) \subseteq \Sigma$. Now, the polygon I_1 is adjacent to Γ, more particularly, it shares one frontier-arc with Γ and the remaining frontier-arcs are shared with polygons lying outside $\Gamma \cup I_2$. Then, $\pi(I_1)$ is not necessarily a polygon of the rigid structure. However, since π is rigid outside $\Gamma \cup I_1$, it respects the adjacency of polygons outside $\Gamma \cup I_1$. Hence, among the $\sharp\,\mathrm{Fr}(I_1)$ images of the frontier-arcs of I_1 by π, $\sharp\,\mathrm{Fr}(I_1) - 1$ are shared with polygons lying outside $\pi(\Gamma \cup I_2) = \Sigma \cup H_3$, whereas one image of a frontier arc is shared with $\pi(\Gamma) \subseteq \Sigma \cup H_1$. See an example of this situation in Figure 25. □

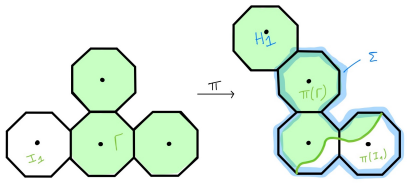


Figure 25

Lemma 3.18. *Consider a* 3*-corner in* $\mathscr{C}(A_{n,m})$ *and assume that we are in one of the following situations.*

- *Let* $n > 1$, $m \geq 1$ *and the* 3*-corner has an attracting root of Type A, B, or D.*
- *Let* $n > 1$ *and* $1 \leq m \leq n+1$ *and the* 3*-corner has an attracting root of Type C.*

Then, $h(\Sigma) \geq 2$*, where* $h(\Sigma)$ *is the height of* Σ*.*

Proof. Recall that, by Lemma 3.17, $\pi(I_1)$ lies in Σ. This means that Σ has enough frontier-arcs so that $\pi(I_1)$ shares its boundary with $\sharp\,\mathrm{Fr}(I_1) - 1$ polygons lying outside $\Sigma \cup H_1$ and $H_3 = \pi(I_2)$ (since I_1 and I_2 are not adjacent). That is, the number of frontier arcs of Σ minus two (for the one shared with H_1 and H_3) must be greater than (or equal to) $\sharp\,\mathrm{Fr}(I_1) - 1$:

$$\sharp\,\mathrm{Fr}(\Sigma) - 2 \geq \sharp\,\mathrm{Fr}(I_1) - 1. \tag{1}$$

We treat each case one by one.

(A) Let $n > 1$, $m \geq 1$, assume that Δ, Σ and Γ do not contain the central polygon M and the Hs', I's and $J's$ are not M. Observe that it is like working with $\mathscr{C}(A_{n,n+1})$. Hence we are in the same situation as a 3-corner of Figure 24 in $\mathscr{C}(A_{n,n+1})$. Observe that since we work with $A_{n,n+1}$, (1) becomes:

$$h(\Sigma)(n-1) + 2 - 2 \geq n.$$

This inequality is satisfied if and only if $n > 1$ and $h(\Sigma) \geq 2$.

(B) Now, $n > 1$, $m \geq 1$ and assume that Δ, Σ and Γ do not contain the central polygon, denoted by M. Moreover, assume that among the adjacent polygons to Δ, Σ and Γ some are the central polygon M. Without loss of generality, we consider the I's.

- If $I_2 = M$: Recall that π is rigid outside $\Gamma \cup \mathrm{I}_1$ thus we have $H_3 = \pi(I_2) = M$. So, observe that I_1 and H_1 are distinct from M. As previously, (1) gives:

$$\sharp\,\mathrm{Fr}(\Sigma) - 2 = h(\Sigma)(n-1) + 2 - 2 \geq n = \sharp\,\mathrm{Fr}(I_1) - 1,$$

and thus $h(\Sigma) \geq 2$.

- If $I_1 = M$: Recall that $[\Delta \cup J_2, \phi] = [\Gamma \cup I_2, \pi]$ so $[\Delta \cup J_2, \mathrm{id}] = [\Gamma \cup I_2, \phi^{-1}\pi]$ thus $\phi^{-1}\pi$ is rigid outside $\Delta \cup J_2$. Moreover, since $[\Delta \cup J_2 \cup J_3, \mathrm{id}] = [\Gamma \cup I_1 \cup I_2, \phi^{-1}\pi]$ one has $\phi^{-1}\pi(I_1) = J_3 = M$. Then one can proceed as in the latter case to obtain that $h(\Sigma) = h(\Delta) \geq 2$.

(D) Finally, take $n > 1$, $m \geq 1$ and assume that Σ contains the central polygon M but Δ and Γ do not. Let $k := h(\Sigma) = h(\Gamma) = h(\Delta)$. Since $\pi(\Gamma \cup I_1) = \Sigma \cup H_1$ and π is rigid outside $\Gamma \cup I_1$ one has $\sharp \operatorname{Fr}(\Gamma \cup I_1) = \sharp \operatorname{Fr}(\Sigma \cup H_1)$. That is,

$$\left.\begin{array}{ll} (k+1)(n-1)+2 & \text{if } I_1 = M \\ m + k(n-1) & \text{if } I_1 \neq M \end{array}\right\} = (k+1)(n-1)+2.$$

On one hand, if $I_1 \neq M$, the previous equality implies that $m = n+1$ so we are in the case of $\mathscr{C}(A_{n,n+1})$, i.e. in situation (A). On the other hand, assume $I_1 = M$. We also let $J_3 = M$ since otherwise by a similar reasoning with $\phi(\Delta \cup J_3) = \Sigma \cup H_3$, we have $m = n+1$ and we already treated this case. By contradiction, assume that $k = 1$ and thus $\Sigma = M$. Recall that $\pi(I_1) \subseteq \Sigma$, π is rigid outside $\Gamma \cup H_1$, $\phi(J_3) \subseteq \Sigma$ and ϕ is rigid outside $\Delta \cup J_3$. Hence, on one side $\pi(I_1) = M$ and $\phi(J_3) = M$. But on the other side this implies that $\pi(I_1)$ is adjacent to $\pi(I_2) = H_3$ and similarly, $\phi(J_3)$ is adjacent to $\phi(J_2) = H_1$, which contradicts the rigidity of π and ϕ at I_2 and J_2, respectively. We conclude that $h(\Sigma) \geq 2$.

(C) Let $n > 1$ and $1 \leq m \leq n+1$. Assume that Δ, Σ and Γ contain the central polygon. By Lemma 3.17 and (1) one has,

$$\sharp \operatorname{Fr}(\Sigma) - 2 = m + (h(\Sigma)-1)(n-1) - 2 \geq \sharp \operatorname{Fr}(I_1) - 1 = n,$$

i.e, $m - 1 \geq 2n - h(\Sigma)(n-1)$. Recall that $1 \leq m \leq n+1$, hence $h(\Sigma) \geq 2$.

□

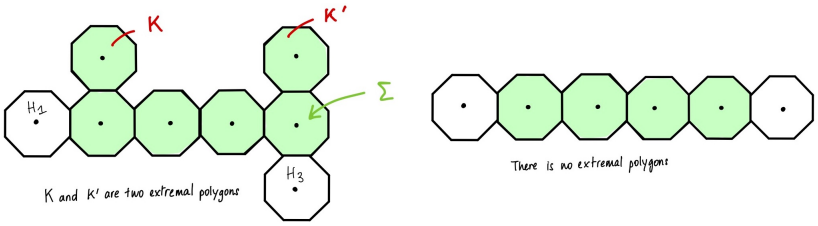


Figure 26

Definition 3.19. Let $n, m \geq 1$ and Σ be an admissible surface in $\mathscr{C}(A_{n,m})$ with two adjacent polygons H_1 and H_3. A polygon $K \subseteq \Sigma$ is an *extremal polygon of the surface* $\Sigma \cup H_1 \cup H_3$ if it shares only one frontier-arc with $\Sigma - K$ and none with H_1 and H_3. See Figure 26 for an illustration.

Lemma 3.20. *Let $n > 1, m \geq 1$ and Σ be an admissible surface in $\mathscr{C}(A_{n,m})$ with $h(\Sigma) \geq 2$. Let H_1 and H_3 be two adjacent polygons to Σ. Let $D \subseteq \Sigma$ be a one-punctured disk such that*

- *$\Sigma - D$ is connected,*
- *D has the same number of frontier-arcs as polygons of the rigid structure (so either $n+1$ or m),*

- *among the frontier arcs of D, one is shared with $\Sigma - D$ and all the others are shared with polygons of the rigid structure lying outside Σ and distinct from H_1 and H_3,*
- *if D has m frontiers arcs then either $m = n + 1$ or $\Sigma \cup H_1$ must contain the central polygon.*

Then, there exist $\chi \in \mathfrak{mod}(A_{n,m})$, an admissible surface Ω and two adjacent polygons H_1' and H_3' of Ω such that

- $[\Omega \cup H_1' \cup H_3', \mathrm{id}] = [\Sigma \cup H_1 \cup H_3, \chi]$,
- $\chi(H_i) = H_i'$, $i = 1, 3$
- $\chi(D) \subseteq \Omega$ *is an extremal polygon in* $\Omega \cup H_1' \cup H_3'$.

Proof. First **assume that there exists an extremal polygon K in $\Sigma \cup H_1 \cup H_3$.**

- If K is not the central polygon and D has n frontier-arcs. By the assumptions in the statement, Σ has n consecutive frontier-arcs $e_1, ..., e_n$ (the one of D) shared with polygons lying outside Σ and distinct from H_1 and H_3. Now $K \subseteq \Sigma$ is an extremal polygon distinct from the central so it has n consecutive edges belonging to the frontier of Σ and not shared with H_1 and H_3. Let $\psi \in \mathfrak{mod}(A_{n,m})$ that shifts cyclically the frontier-arcs of Σ such that $e_1, ..., e_n$ are mapped to the frontier-arcs of K. See Figure 27

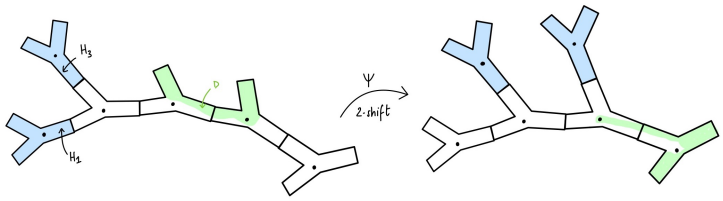


Figure 27

Let $H_1' = \psi(H_1)$ and $H_3' = \psi(H_3)$. Then it remains to untwist $\psi(D)$ in

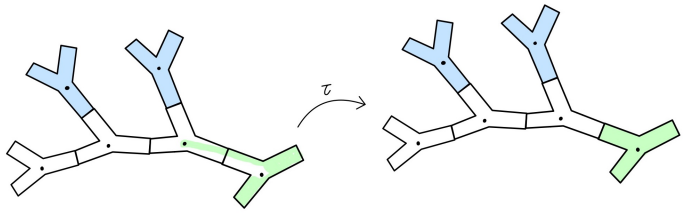


Figure 28

K with a braided element (i.e. a half-Dehn twist) $\tau \in \mathfrak{mod}(A_{n,m})$ as in Figure 28. Then let $\chi = \tau \circ \psi$, $\Omega = \chi(\Sigma)$, $H_1' = \chi(H_1)$ and $H_3' = \chi(H_3)$. By construction $[\Omega \cup H_1' \cup H_3', \mathrm{id}] = [\Sigma \cup H_1 \cup H_3, \chi]$, and $\chi(D) \subseteq \Omega$ is an extremal polygon in $\Omega \cup H_1' \cup H_3'$.

- If K is the central polygon and D has m frontier-arcs. Here the proof is the same as above by replacing n with $m - 1$.

- If K is the central polygon and D has n frontier-arcs. Let $R_1, ...R_m$ be the m strands of polygons starting out the central polygon K. Let M_i be the polygon in R_i adjacent to K. Without loss of generality let us say that R_1 is the string containing $\Sigma - K$. Take

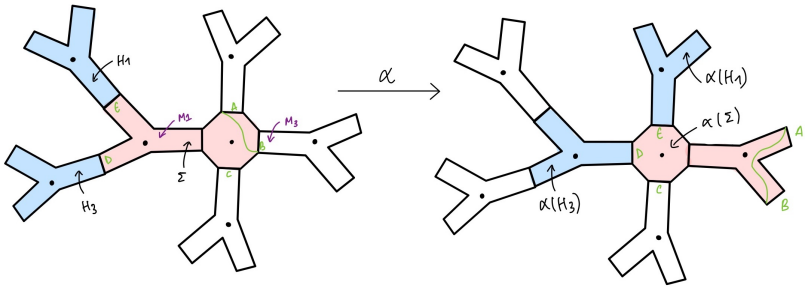


Figure 29

  $\alpha \in \mathfrak{mod}(A_{n,m})$ such that α pushes R_1 into R_3 as in Figure 29. Then $\alpha(K \cup M_1) = K \cup M_3$ and α is rigid outside $K \cup M_1$. Let $\Sigma' = \alpha(\Sigma)$, $N_1 = \alpha(H_1)$ and $N_3 = \alpha(H_3)$. Then $[\Sigma' \cup N_1 \cup N_3, \mathrm{id}] = [\Sigma \cup H_1 \cup H_3, \alpha]$ and Σ' has an extremal polygon that is not the central one. Hence on can apply the same reasoning as in the first case with $[\Sigma' \cup N_1 \cup N_3, \mathrm{id}]$ and $\alpha(D)$ to conclude.

- if K is not the central polygon and D has m frontier-arcs. By the last assumption of the statement, $\Sigma \cup H_1$ contains the central polygon or $m = n + 1$.
  - Assume that $\Sigma \cup H_1$ contains the central polygon. Then $\Sigma \cup H_1 \cup H_3$ has the same number of punctures, the same number of boundary components and so the same euler characteristic as a surface, with height $h(\Sigma) + 2$, containing the central polygon as an extremal polygon. Thus we are in the same situation as at the beginning of the proof, if K is the central polygon and D has m frontier-arcs.
  - Now assume that $m = n + 1$ then D has $n + 1$ frontier arcs, we do not distinguish a central polygon, we already treated this case.

Then, **assume that there exists no extremal polygon in** $\Sigma \cup H_1 \cup H_3$. Observe that,

Fact 3.21. The two distinct polygons in Σ adjacent to H_1 and H_3, denoted B_1 and B_2, share only one frontier-arc with $\Sigma - B_1$ resp. $\Sigma - B_2$.

First assume that D has n frontier arcs. Recall that since D lies in Σ so the latter surface has at least n consecutive frontiers arcs shared with polygons distinct from H_1 and H_3 and lying outside Σ. Thus one can shift cyclically the frontier arcs of Σ such that one of the B_i becomes extremal. This is possible because we need to apply at most a $\frac{n}{2}$-shift and one has at least n consecutive frontiers arcs shared with polygons distinct from H_1 and H_3. See Figure 30. Denote by ψ the cyclic shift that is used, then $[\Sigma \cup H_1 \cup H_3, \psi] = [\Sigma \cup N_1 \cup N_3, \mathrm{id}] =$ where $N_1 = \psi(H_1)$ and $N_3 = \psi(H_3)$. Since $\Sigma \cup N_1 \cup N_3$ has an extremal polygon we can apply the result above to conclude.

Finally, assume that D has m frontier arcs. By the assumption $\Sigma \cup H_1$ contains the central polygon or $m = n + 1$. Assume that $\Sigma \cup H_1$ contains the central polygon. Then $\Sigma \cup H_1 \cup H_3$ as the same number of punctures, the same number of boundary components and so the same

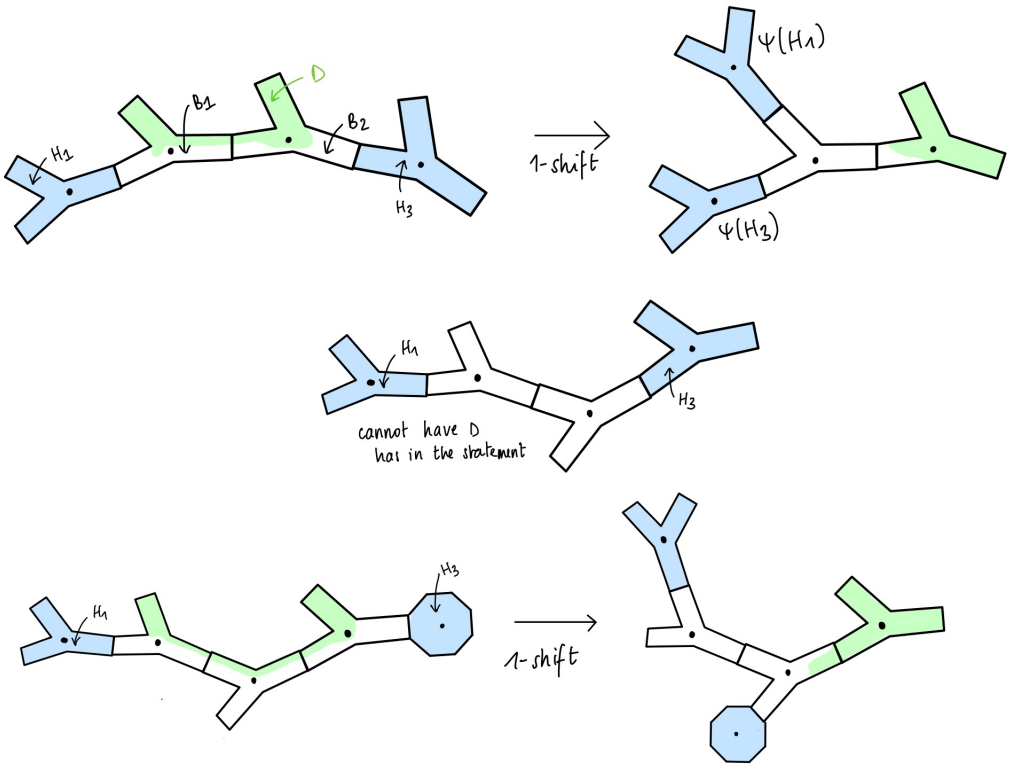

D
B1
B2
H2
H3
1-shift
Ψ(H1)
Ψ(H3)
H1
H3
cannot have D
has in the statement
H1
H3
1-shift

Figure 30

euler characteristic than a surface, with height $h(\Sigma)+2$, containing the central polygon as an extremal polygon.. Thus we are in the same situation as at the beginning of the proof, if K is the central polygon and D has m frontier-arcs. Now, if $m = n+1$, then we are in the case where D has $n+1$ frontier-arcs and we already treated this case. □

Proof of Proposition 3.14. Consider a 3-corner in $\mathscr{C}(A_{n,m})$ and assume that the assumptions in the statement of Proposition 3.14 are satisfied. That is,

- either $n > 1$, $m \geq 1$ and the 3-corner has an attracting root of Type A, B, or D,
- or, $n > 1$ and $1 \leq m \leq n+1$ and the 3-corner has an attracting root of Type C.

By Lemma 3.18 $h(\Sigma) \geq 2$ one can apply Lemma 3.20 with $D = \pi(I_1)$. Indeed $\pi(I_1)$ satisfies the assumptions of Lemma 3.20. In particular, if I_1 is the central polygon and $\Sigma \cup H_1$ does not contain the the central polygon, then (i) gives $\mathrm{Fr}(\pi(\Gamma \cup I_1)) = \mathrm{Fr}(\Sigma \cup H_1)$ i.e. $m + k(n-1) = (k+1)(n-1) + 2$ that is $m = n+1$.

Hence we rewrite:

$$[\Sigma \cup H_1 \cup H_3, \mathrm{id}] = [\Omega \cup H_1' \cup H_3', \chi^{-1}] = [\Omega - K \cup K \cup H_1' \cup H_3', \chi^{-1}],$$

where K is a polygon of the rigid structure. Thus, one can complete our 3-corner into a 3-cube by adding the vertex $[\Omega - K, \psi]$. See Figure 31.

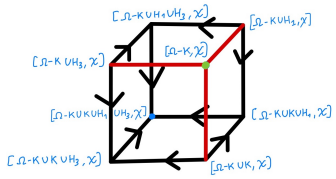


Figure 31

□

Proof of Proposition 3.15. The second part of the Proposition already holds by Remark 3.16 and Proposition 3.14. Hence it remains to show the first part.

Let $n > 1, m \geq 1$ and $m \neq n+1$. We consider the situation in Figure 24. Assume that Σ does not contain M but Δ and Γ do. We want to show that we cannot have such a 3-corner. Since $\pi(\Gamma \cup I_1) = \Sigma \cup H_1$, π is rigid outside $\Gamma \cup I_1$ (i), $\phi(\Delta \cup J_3) = \Sigma \cup H_3$ and ϕ is rigid outside $\Delta \cup J_3$ (ii), we must have:

- $\sharp\,\mathrm{Fr}(\Gamma \cup I_1) = \sharp\,\mathrm{Fr}(\Sigma \cup H_1)$, i.e. $m + k(n-1) = \begin{cases} (k+1)(n-1)+2 & \text{if } H_1 \neq M \\ m + k(n-1) & \text{if } H_1 = M \end{cases}$
- $\sharp\,\mathrm{Fr}(\Delta \cup J_3) = \sharp\,\mathrm{Fr}(\Sigma \cup H_3)$, i.e. $m + k(n-1) = \begin{cases} (k+1)(n-1)+2 & \text{if } H_3 \neq M \\ m + k(n-1) & \text{if } H_3 = M \end{cases}$

So, if $H_1 \neq M$ or $H_3 \neq M$ one has $m = n+1$, however, we assumed $m \neq n+1$. Now, if $H_1 = H_3 = M$ one has $\pi(I_2) = H_3 = M$ and $\phi(J_2) = H_1 = M$. The rigidity of π and ϕ gives $I_2 = J_2 = M$. Thus, $\pi(\Gamma \cup I_2) = \phi(\Delta \cup J_2)$. Since Γ and Δ contain M this is equivalent to $\pi(\Gamma) = \phi(\Delta)$ hence $[\Gamma, \pi] = [\Delta, \phi]$ and we do not have a 3-corner. □

Proof of Proposition 3.12. Proposition 3.12 follows directly from the three Propositions 3.15, 3.14 and 3.15. □

3.3 When is $\mathscr{C}(A_{n,m})$ a CAT(0) cube complex?

In this section, we determine for which sub-family of $A_{n,m}$ the associated cube $\mathscr{C}(A_{n,m})$ is CAT(0).

Example 3.22. We show that $\mathscr{C}(A_{1,2})$ is CAT(0). By Lemma 3.5 it is enough to show that $\mathscr{C}(A_{1,2})$ satisfies the assumptions in Proposition 2.28. By Theorem 3.3 in [GLU22], $\mathscr{C}(A_{1,2})$ is simply connected. Then, by Lemma 3.8, there is no copy of $K_{2,3}$. Moreover by Lemma 3.7 we do not have 1-corners. Now, by Lemma 3.10 it only remains to show that in $\mathscr{C}(A_{1,2})$ every 3-corner with an attracting root can be completed into a 3-cube. This follows from the fact that $\mathscr{C}(A_{1,2})$ is a square-complex (there is only two ways to remove or add adjacent polygons). Indeed, consider the general situation in Figure 24. Then, since $\pi(I_2) = H_3$ and π is rigid outside $\Gamma \cup \mathrm{I}_1$ one cannot have $\pi(I_1) \subseteq \Sigma$. See Figure 32. Thus, $\pi(I_1) = H_1$ and

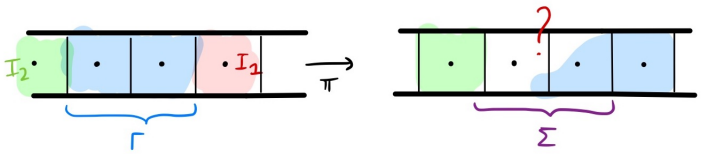


Figure 32

so $\pi(\Gamma) = \Sigma$. Hence, $[\Sigma, \mathrm{id}] = [\Gamma, \pi]$. Consequently, our 3-corner is in fact a 2-cube. And we conclude that $\mathscr{C}(A_{1,2})$ is CAT(0). Similarly one can show that $\mathscr{C}(A_{1,1})$ is CAT(0). △

Counter-example 3.23. The cube complex $\mathscr{C}(A_{1,m})$ where $m \geq 3$ is not CAT(0). In Section 3.3 of the article [GLU22], the authors give a counter-example for the case $A_{1,3}$. We can extend it to the cases $A_{1,m}$ with $m \geq 3$. Indeed, consider the situation illustrated in Figure 33. Then,

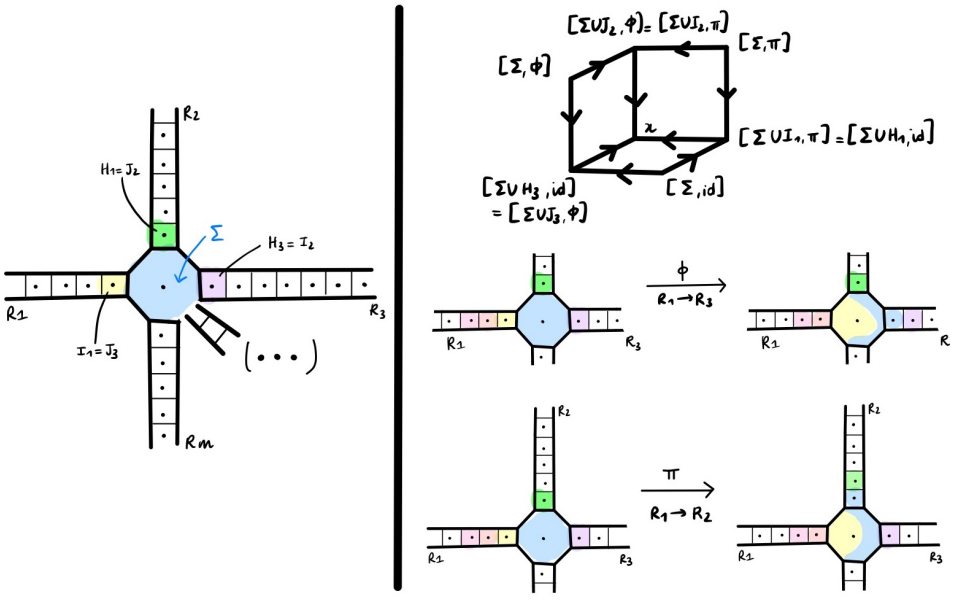


Figure 33

one cannot complete the 3-corner above into a 3-cube because there is no vertex of height zero. Thus, we need to modify the definition our cube $\mathscr{C}(A)$ to get a CAT(0) cube complex associated to $\mathfrak{mod}(A_{1,m})$ for $m \geq 3$.

Counter-example 3.24. The cube complex $\mathscr{C}(A_{n,m})$ is not CAT(0) for all $m > n+1$. Consider the 3-corner in $\mathscr{C}(A_{n,m})$ with the same orientation as in Figure 24 and with the following vertices. Let Σ, Δ and Γ be of height one and equal to the central polygon, denoted by M. Denote by $R_1, ...R_m$ the m strands that start out the central polygon. An *infinite line of polygons* L in R_j is an infinite chain of polygons $A^1, A^2, ..$ starting from the polygon P_j in R_j adjacent to M and such that A^k and A^{k+1} are adjacent for all $k \geq 1$.

Fix three *infinite lines* of polygons L_1, L_2 and L_3 in R_1, R_2 and R_3. The line L_i is built recursively as follows, start from P_i then you add the adjacent polygon whose puncture corresponds to the right-most children of the vertex corresponding to the puncture of P_i. Then we iterate the process by starting with the polygon that we just added.

Let π be an asymptotically rigid homeomorphism that shifts each polygon of $L_1 \cup L_2$ by one polygon such that $\pi(A_1^1)$ is in the central polygon. Let ϕ be an asymptotically rigid homeomorphism that shifts each polygon of $L_1 \cup L_3$ by one polygon such that $\phi(A_1^1)$ is the central polygon. Let $H_1 = J_2$ (resp. $I_1 = J_3$, $H_3 = I_2$) be the polygon in R_2 (resp. R_1, R_3) adjacent to the central polygon. See Figure 34 for an illustration with $m = 4$ and $n = 2$.

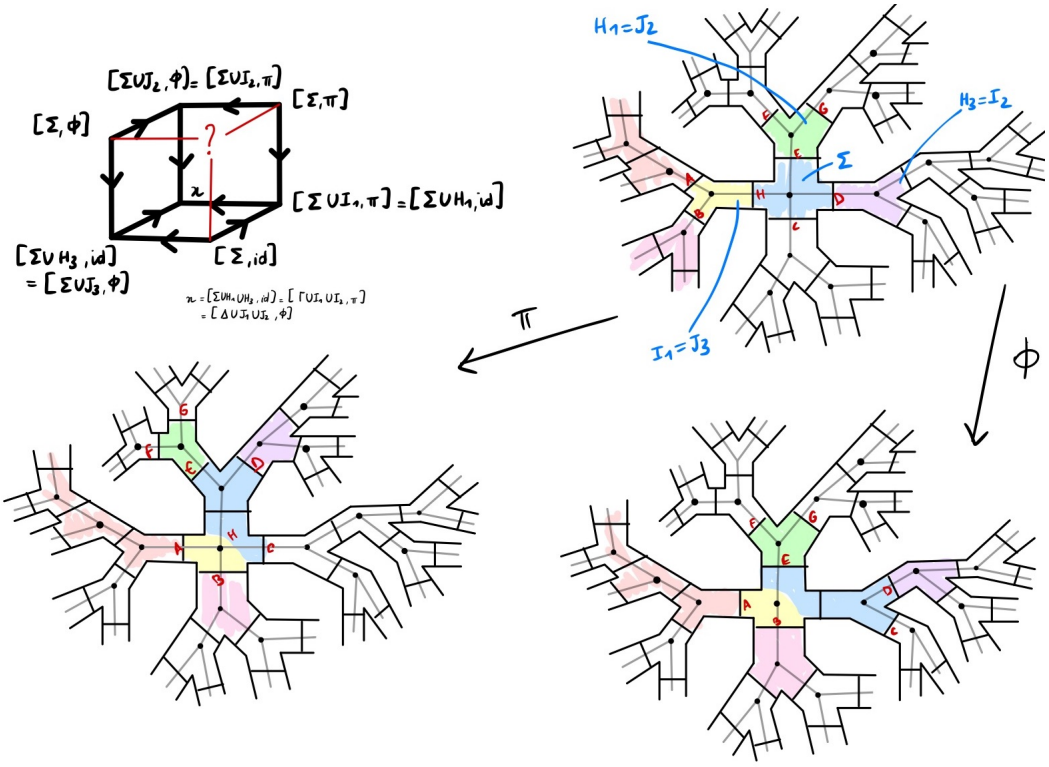


Figure 34

Then by construction, we cannot complete this 3-corner into a 3-cube because we do not have a vertex of height zero.

Theorem 3.25. *Let $n \geq 1$, the cube complex $\mathscr{C}(A_{n,m})$ is* CAT(0) *if and only if $1 \leq m \leq n+1$.*

Proof. Assume that $n \geq 1$ and $1 \leq m \leq n+1$. The case $n = 1$ is given by the latter Example 3.22. Now assume that $n > 1$ and $1 \leq m \leq n+1$. The underlying graph of $\mathscr{C}(A_{n,m})$ is clearly connected. Proposition 3.5 allows us to use Proposition 2.28 with $\mathscr{C}(A_{n,m})$ instead of $A'^{\square}_{n,m}$ in order to prove Theorem 3.25. Thus, we need to show that $\mathscr{C}(A_{n,m})$ satisfies the assumptions in Proposition 2.28. By Theorem 3.3 in [GLU22], $\mathscr{C}(A_{n,m})$ is contractible and consequently simply connected. Moreover, by Lemma 3.7 we do not have 1-corner. Then, every 3-corner is of type $(3,0)$, $(2,1)$, $(1,2)$ or $(0,3)$. By Lemma 3.10 and Proposition 3.12 every 3-corner in $\mathscr{C}(A_{n,m})$ can be completed into a 3-cube. To finish the proof, it remains to show that $\mathscr{C}(A_{n,m})$ does not contain a copy of the bipartite graph $K_{2,3}$ this is exactly the content of Lemma 3.8.

For the converse, we prove the contrapositive. Let $m > n+1$, if $n = 1$ by Counter-example 3.23 $\mathscr{C}(A_{1,m})$ is not CAT(0) for $m \geq 3$. Then if $n > 1, m > n+1$, $\mathscr{C}(A_{n,m})$ is not CAT(0) by Counter-example 3.24.

This concludes the proof of Theorem 3.25. □

Remark 3.26. Let $m, n \geq 1$, the subcomplex of $\mathscr{C}(A_{n,m})$ generated by the vertices of the form $[\Sigma, \phi]$ where Σ is an admissible surface containing the central polygon is called the *spine* and is denoted by $\mathscr{SC}(A_{n,m})$. Observe that the previous Counter-examples 3.23 and 3.24 work with the spine. Moreover, the spine is contractible by Theorem 3.3 and Proposition 5.2 in [GLU22]. Now, the proof that the spine do not have 1-corners and 2-corner is identical to the one for $\mathscr{C}(A_{n,m})$ (see 3.3 and 3.8). Observe that a 3-corner in the spine is always of Type (C) 3.13.

Hence by the previous and Proposition 3.14 we deduce that $\mathscr{SC}(A_{n,m})$ is CAT(0) if and only if $n \geq 1$ and $1 \leq m \leq n+1$.

4 Studying another CAT(0) cube complex $\mathscr{D}(A_{n,m})$

In this section, we modify the definition of $\mathscr{C}(A_{n,m})$ to get a cube complex that is CAT(0) for all $n \geq 3, m \geq 1$. Finally, we exploit the CAT(0) property of our cube complex to deduce properties about $\mathfrak{mod}(A_{n,m})$.

4.1 Looking for another CAT(0) cube complex

Consider a locally finite tree $A_{n,m}$, and take its arboreal surface $\mathscr{S}(A_{n,m})$. Let $\mathscr{S}^*(A_{n,m})$ denote the punctured arboreal surface obtained from $\mathscr{S}(A_{n,m})$ by adding a puncture for each vertex of the tree excepting the vertex of valence m. We fix a rigid structure on $\mathscr{S}^*(A_{n,m})$. To this end, we divide the surface into polygons with a family of pairwise non-intersecting arcs whose endpoints are on the boundary of the surface such that:

- each arc crosses once and transversely a unique edge of the tree
- each arc contains exactly one vertex of the underlying tree in its interior.

Then, one reformulates our definitions with respect to this new structure.

Let A be a locally finite tree, a subsurface $\Sigma \subseteq \mathscr{S}^*(A)$ is said *admissible* if it can be written as the finite union of polygons from the rigid structure. The *height* of Σ, denoted $h(\Sigma)$, is the number of punctures inside Σ. The central polygon has height zero.

An *asymptotically rigid homeomorphism* of $\mathscr{S}^*(A)$ is an element ϕ from $\mathscr{S}^*(A)$ to itself which respects the rigid structure almost everywhere. That is, there exists an admissible surface called the *support* of ϕ, denoted $\operatorname{supp}^* \phi$ such that:

- its image $\phi(\Sigma)$ is an admissible surface,
- outside Σ, the map ϕ sends polygons to polygons, i.e. $\phi : \mathscr{S}^*(A) - \Sigma \longrightarrow \mathscr{S}^* - \phi(\Sigma)$ is *rigid*.

We call $\mathfrak{mod}(\mathscr{S}^*(A)$ the group of isotopy classes of orientation-preserving homeomorphisms of $\mathscr{S}^*(A)$.

Lemma 4.1. *We have an isomorphism*

$$\mathfrak{mod}(\mathscr{S}^*(A_{n,m})) \cong \mathfrak{mod}(\mathscr{S}^\sharp(A_{n,m}))$$

Proof. Fix an infinite line of polygons $L_1 = A_1^1, A_1^2 ...$ in R_1. Take the homeomorphism $\psi : \mathscr{S}^\sharp(A_{n,m}) \longrightarrow \mathscr{S}^*(A_{n,m})$ that shift the central polygon into A_1^1 and then A_1^1 into A_1^2 and so on. See Figure 35 for the case $n = 1$. If $n > 1$ we can take ψ described in Figure 36. Thus, $\operatorname{Mod}(\mathscr{S}^\sharp(A_{n,m})) \cong \operatorname{Mod}(\mathscr{S}^*(A_{n,m}))$. Now, the homeomorphism ψ sends each polygon of $\mathscr{S}^\sharp(A_{n,m})$ to a polygon of $\mathscr{S}^*(A_{n,m})$ with the exception of one, which is sent to the union of two polygons. Hence, the conjugation by ψ gives an isomorphism $\mathfrak{mod}(\mathscr{S}^*(A_{n,m})) \cong \mathfrak{mod}(\mathscr{S}^\sharp(A_{n,m}))$. □

Let $\phi \in \mathfrak{mod}(\mathscr{S}^\sharp(A_{n,m}))$, one observes that $\operatorname{supp}^* \phi$ is different from $\operatorname{supp} \psi\phi\psi^{-1}$.

Definition 4.2. Let $\mathscr{D}(A_{n,m})$ be the cube complex given by:

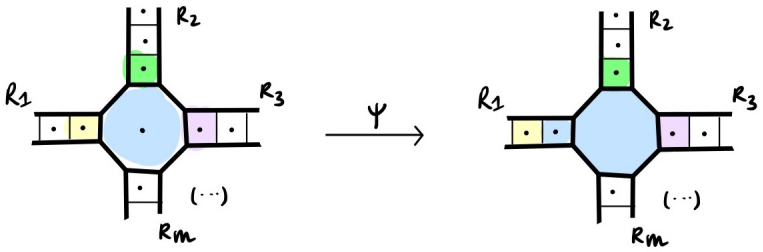


Figure 35

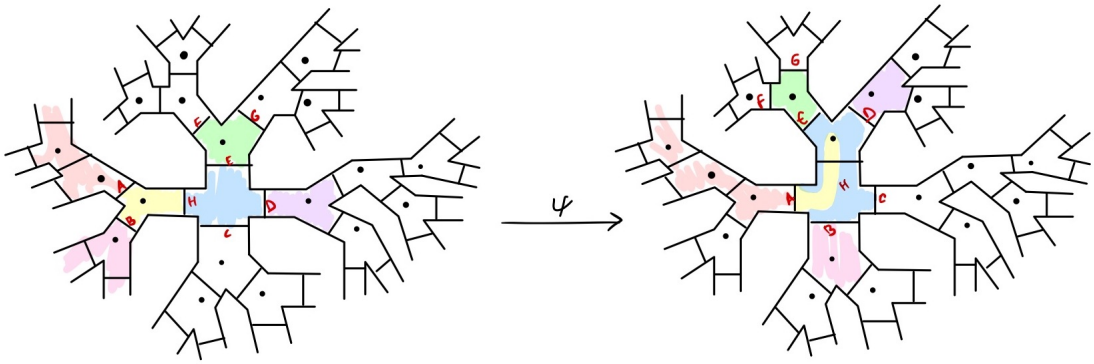


Figure 36

- vertices: the equivalence classes $[\Sigma, \phi]$ with the same equivalence relation as in $\mathscr{C}(A_{n,m})$ with $\Sigma \subseteq \mathscr{S}^*(A_{n,m})$ an admissible surface, with respect to the rigid structure described above, that contains the central polygon.
- edges: We have edges between $[\Sigma, \phi]$ and $[\Sigma \cup H, \phi]$ with H an adjacent polygon to Σ.
- cubes: Take a vertex $[\Sigma, \phi]$, and distinct adjacent polygons $H_1, ..., H_k$ to Σ. Then the subgraph $\left\{[\Sigma \cup \bigcup_{i \in I} H_i, \phi] | I \subset \{1, ..., k\}\right\}$ is filled with a k-cube.

Definition 4.3. Let x be a vertex in $\mathscr{D}(A_{n,m})$ and $\Sigma \subseteq \mathscr{S}^*(A_{n,m})$ be an admissible surface. If there exists a finite path of vertices $x_0, x_1, ...x_n$ from $x_0 = x$ to $x_n = [\Sigma, \mathrm{id}]$ such that $h(x_i) > h(x_{i-1})$, one says that $[\Sigma, \mathrm{id}]$ *dominates* x.

Definition 4.4. Let $\mathcal{S}$ be a finite collection of vertices in $\mathscr{D}(A_{n,m})$. We denote by $X(\mathcal{S}, \Sigma)$ the subcomplex in $\mathscr{D}(A_{n,m})$ generated by the height-increasing paths from a vertex in $\mathcal{S}$ to $[\Sigma, \mathrm{id}]$.

Proposition 4.5. *Let $m \geq 3, n \geq 1$ then the cube complex $\mathscr{D}(A_{n,m})$ is* CAT(0).

Remark 4.6. $\mathscr{D}(A_{1,2})$ and $\mathscr{D}(A_{1,1})$ are also CAT(0). See Example 3.22.

Before proceeding to the proof of the proposition we need to verify two technical lemmata. We follow closely the proof of the contractility of $\mathscr{C}(A_{n,m})$ in Section 3.2 [GLU22].

Lemma 4.7. *Let $\mathcal{S}$ be a finite collection of vertices in $\mathscr{D}(A_{n,m})$. Then, there exists an admissible surface Σ such that $[\Sigma, \mathrm{id}]$ dominates all the vertices in $\mathcal{S}$.*

Proof. We denote by $([\Delta_i, \phi_i])_{i\in I}$ the collection of vertices in $\mathcal{S}$. Let Σ_i be the admissible surface supporting ϕ_i for each $i \in I$. we denote by Ω_i the smallest admissible surface such that $\Sigma_i \cup \Delta_i \subseteq \Omega_i$, for each $i \in I$. Then let Σ be an admissible surface containing all the $\phi(\Omega_i)$. One obtains a height-increasing path from $[\Delta_i, \phi_i]$ to $[\Omega_i, \phi_i]$ by adding polygons. Similarly, by definition of Σ, one has a second height-increasing path from $[\Omega_i, \phi_i]$ to $[\Sigma, \mathrm{id}]$. By concatenating the two paths we obtain a height-increasing path from $[\Delta_i, \phi_i]$ to $[\Sigma, \mathrm{id}]$. Thus $[\Sigma, \mathrm{id}]$ dominates all the vertices in $\mathcal{S}$. □

Lemma 4.8. *Let $\mathcal{S}$ be a finite collection of vertices in $\mathscr{D}(A_{n,m})$. Then, there exists an admissible surface Σ such that $X(\mathcal{S}, \Sigma)$ is contractible.*

Proof. Let $\mathcal{S}$ be a finite collection of vertices in $\mathscr{D}(A_{n,m})$. By Lemma 4.7 there exists an admissible surface Σ such that $[\Sigma, \mathrm{id}]$ dominates all the vertices in $\mathcal{S}$. If $X(\mathcal{S}, \Sigma) = [\Sigma, \mathrm{id}]$ we are done. Assume that $X(\mathcal{S}, \Sigma) \neq [\Sigma, \mathrm{id}]$. Now, take a vertex $x \in X$ with minimal height. Then, since $X(\mathcal{S}, \Sigma) \neq [\Sigma, \mathrm{id}]$, x admits $x_1, ..., x_k$ as neighbours in $X(\mathcal{S}, \Sigma)$. By Lemma 3.3 one can label $x = [\Delta, \phi]$ and $x_i = [\Delta \cup H_i, \phi]$ for $1 \leq i \leq k$ where Δ is an admissible surface and $H_1, ..., H_k$ are some adjacent polygons.

Claim 4.9. The cube generated by the vertices $\{[\Delta \cup_{i\in I} H_i | I \subseteq \{1, ..., k\}]\}$ lies in $X(\mathcal{S}, \Sigma)$

Proof. One has a height-increasing path from x to $[\Sigma, \mathrm{id}]$. Thus by Lemma 3.3 $[\Sigma, \mathrm{id}] = [\Delta \cup J_1 \cup ... \cup J_n, \phi]$ where $J_1, ..., J_n$ are some polygons. Similarly since one also has an increasing path from $x_i = [\Delta \cup H_i, \phi]$ to $[\Sigma, \mathrm{id}]$, by Lemma 3.3, $[\Sigma, \mathrm{id}] = [\Delta \cup H_i \cup M_1 \cup ... \cup M_l, \phi]$ where $M_1, ..., M_l$ are some polygons. To sum up:

$$[\Sigma, \mathrm{id}] = [\Delta \cup J_1 \cup ... \cup J_n, \phi] = [\Delta \cup H_i \cup M_1 \cup ... \cup M_l, \phi],$$

so $l = n$ and $J_i = M_{\sigma(i)}$ for some permutation σ. Hence, one can add to Δ the adjacent polygons H_i and next the remaining J_i to obtain a path from x to $[\Sigma, \mathrm{id}]$ passing through the $\{[\Delta \cup_{i\in I} H_i | I \subseteq \{1, ..., k\}]\}$. □

In particular, all the direct neighbours x_i's of x in $X(\mathcal{S}, \Sigma)$ are contained in a cube in $X(\mathcal{S}, \Sigma)$. Thus the complex $X(\mathcal{S}, \Sigma)$ deformation retracts onto $X((\mathcal{S} - \{x\}) \cup \{x_1, ..., x_k\}, \Sigma)$. Since $[\Sigma, \mathrm{id}]$ dominates $(\mathcal{S} - \{x\}) \cup \{x_1, ..., x_k\}$ we repeat the process. Hence we find a sequence:

$$X(\mathcal{S}, \Sigma) \supset X(\mathcal{S}_1, \Sigma) \supset ...$$

This sequence stop since $X(\mathcal{S}, \Sigma)$ contains finitely many cells. Hence we obtain from a certain rank k, $X(\mathcal{S}_k, \Sigma) = [\Sigma, \mathrm{id}]$ for some finite collection $\mathcal{S}_k$ dominated by $[\Sigma, \mathrm{id}]$. Thus $X(\mathcal{S}, \Sigma)$ deformation retracts onto $[\Sigma, \mathrm{id}]$ so is contractible

□

Proposition 4.10. *Let $m, n \geq 1$ the cube complexe $\mathscr{D}(A_{n,m})$ is contractible.*

Proof. Let $n \geq 1$ and $\psi : \mathbb{S}^n \longrightarrow \mathscr{D}(A_{n,m})$ be a continuous map. The image of ψ lies in compact subcomplex which has a finite collection of vertices $\mathcal{S}$. By Lemma 4.7, there exists an admissible surface Σ such that $[\Sigma, \mathrm{id}]$ dominates all the vertices in $\mathcal{S}$. Thus the image of ψ lies in $X(\mathcal{S}, \Sigma)$ which is contractible by Lemma 4.8. Hence ψ is homotopically trivial. By Whitehead's theorem we deduce that $\mathscr{D}(A_{n,m})$ is contractible. □

Proof of Proposition 4.5. Let $m \geq 3, n \geq 1$. The underlying graph of $\mathscr{D}(A_{n,m})$ is connected. By Proposition 3.5 the cube completion of the underlying graph of $\mathscr{D}(A_{n,m})$ equals $\mathscr{D}(A_{n,m})$ so one can use Proposition 2.28 with $\mathscr{D}(A_{n,m})$ to show that $\mathscr{D}(A_{n,m})$ is CAT(0). We show that the latter satisfies the assumptions in the statement of Proposition 2.28.

- First, there is no copy of the bipartite graph $K_{2,3}$ in $\mathscr{D}(A_{n,m})$. The proof is the same as Lemma 3.8.

- Using Lemma 3.3, as in the proof of Lemma 3.8, it is clear that we do not have 1-corners. Then we need to show that $\mathscr{D}(A_{n,m})$ satisfies the 3-cube condition. Recall that a 3-corner is either of type $(3,0)$, $(2,1)$, $(1,2)$ or $(0,3)$. To show that for all $n, m \geq 1$, the 3-corners of type $(3,0)$, $(2,1)$ or $(1,2)$ in $\mathscr{C}(A_{n,m})$ can be completed into a 3-cube the proof is identical to the one of Lemma 3.10. Now it remains to treat the case $(0,3)$. Observe that by definition of $\mathscr{D}(A_{n,m})$ a 3-corner with an attracting root is always of Type (C) 3.13. Recall that for the case $\mathscr{C}(A_{n,m})$, in the proof of Theorem 3.14, the extra assumption that $1 \leq m \leq n+1$ was only needed when we wanted to complete a 3-corner with an attracting root and of type $(0,3)$. Indeed, this assumption gives that the height of the vertices in the 3-corner is at least 2, allowing us to use Lemma 3.18 whose proof is identical for our cube $\mathscr{D}(A_{n,m})$. Hence, if the the height of the vertices in the 3-corner is at least 2 we are done by Lemma 3.18. Then, assume that the height of the vertices in the 3-corner is 1. Consider the general situation in Figure 24 with admissible surface in $\mathscr{D}(A_{n,m})$.

  Claim 4.11. There exist an asymptotically rigid homeomorphism χ, an admissible surface Ω with two adjacent polygons H_1' and H_3' such that:

  - $\chi(H_i') = H_i$ and $\chi(\Omega) = \Sigma$
  - the central polygon M is adjacent to both H_i'
  - $[\Sigma \cup H_1 H_3, \mathrm{id}] = [\Omega \cup H_1' H_3', \chi]$

  Proof. Let $\Sigma = M \cup H$ If the H_i are both adjacent to M we are done. Otherwise,

  - either H_1 and H_3 are not adjacent to M then, they are adjacent to H and lying in the R_1 strand. Since $m \geq 3$ one can push R_1 into R_2 using ψ described in Figure 37. Then let $\chi := \psi^{-1}$, $\Omega := \chi(\Sigma)$ and $H_i' = \chi(H_i)$.

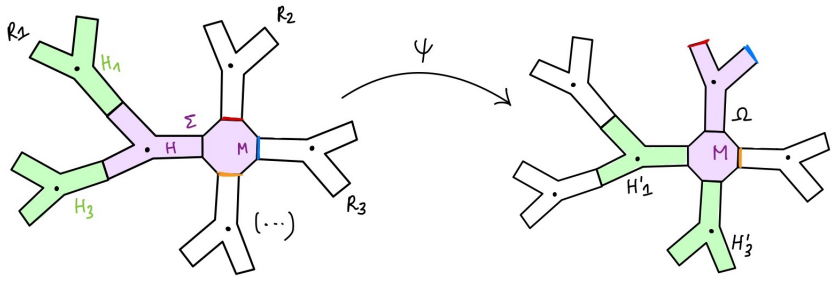


Figure 37

– or H_1 is adjacent to M whereas H_3 is not We proceed as in the previous case but we need to be careful to push the strands where H_3 lies in the one where H_1 does not lie.

□

By the previous claim one can complete our 3-corner into a 3-cube by adding the vertex $[M, \chi]$. See Figure 38.

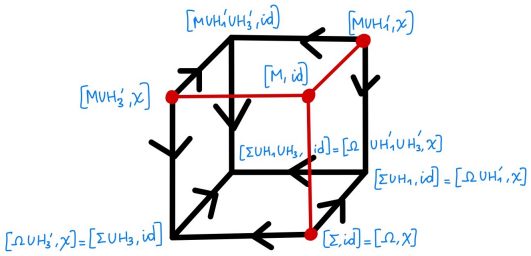


Figure 38

- Finally, $\mathscr{D}(A_{n,m})$ is simply connected by Proposition 4.10.

By Proposition 2.28 $\mathscr{D}(A_{n,m})$ is CAT(0). □

We can exploit the CAT(0)ness of the cube complex $\mathscr{D}(A_{n,m})$. Let $n \geq 1, m \geq 3$ in this section, we deduce some geometric properties of $\mathfrak{mod}(A_{n,m})$ from the fact that it acts on the cube complex $\mathscr{D}(A_{n,m})$ which is CAT(0).

Example 4.12. Consider the element $\beta \in \mathfrak{mod}(A_{1,4})$ described by Figure 39. Let M be the central polygon. Then β fixes the vertex of height zero $[M, \mathrm{id}]$ in $\mathscr{D}(A_{1,4})$.

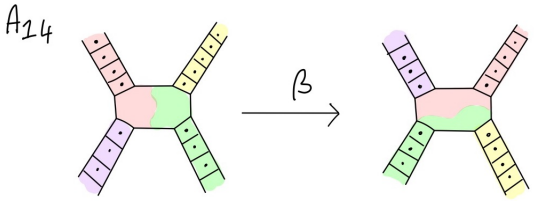


Figure 39

△

Example 4.13. Consider the element $\alpha_1 \in \mathfrak{mod}(A_{1,4})$ described in Figure 40.

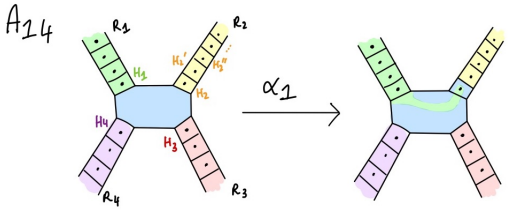


Figure 40

Let M be the central polygon. Then observe that the support $\mathrm{supp}^*\,\alpha_1$ is $M \cup H_1 \cup H_2$. Moreover one has the following equalities:

1. $[M \cup H_1 \cup H_2, \alpha_1] = [M \cup H_2 \cup H_2', \mathrm{id}]$
2. $[M \cup H_2 \cup H_2' \cup H_2'', \mathrm{id}] = [M \cup H_1 \cup H_2 \cup H_2', \alpha_1]$.

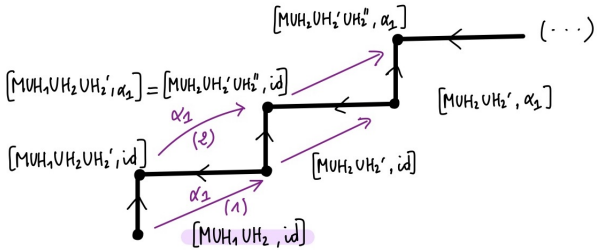


Figure 41

Consequently α_1 fixes the geodesic in $\mathscr{D}(A_{1,4})$ represented in Figure 41 and has translation length two. △

In the previous section we showed that $\mathfrak{mod}(A_{n,m})$ acts on $\mathscr{D}(A_{n,m})$ which is CAT(0) by Theorem 4.5. Because of the definition of the height-orientation on $\mathscr{D}(A_{n,m})$, $\mathfrak{mod}(A_{n,m})$ acts without inversions. Thus, by Theorem 1.4 from [Hag07], we deduce that every element is either elliptic or loxodromic.

Theorem 4.14. *Let $m \geq 3, n \geq 1$, then every element of $\mathfrak{mod}(A_{n,m})$ is either elliptic or loxodromic.*

A next step can be to compute the translation length of any loxodromic element in $\mathfrak{mod}(A_{n,m})$.

5 Finite presentatility of the asymptotically mapping class group associated to $A_{1,m}$

In 2008, Funar and Kaboudjian showed that the braided Ptolemy-Thompson group is finitely presented [FK08]. In this section we start by considering a particular cube complex extracted from the spine $\mathscr{S}(A_{1,m})$ and we show that it is simply connected by following closely Section 5 in [GLU22]. Then, we compute a presentation for $\mathfrak{mod}(A_{1,3})$ by using its action on a simply connected cube complex and the algorithm given by Kenneth Brown in [Bro84]. Then we give a finite presentation for $\mathfrak{mod}(A_{1,3})$. Finally we prove that $\mathfrak{mod}(A_{1,m})$ is finitely presented.

5.1 The height-bounded complex

In this section we follow closely (almost word by word) Section 5 in [GLU22]. The proof is simplified because we are working with $A_{1,m}$ (in particular for the beginning of the proof of Proposition 5.8 in [GLU22]).

Definition 5.1. Let S be a surface with boundary. Let P be a set of punctures and M a set of marked points. The we denote by $\mathcal{R}(S, P, M)$ the simplicial complex defined by:

- vertices: isotopy classes of arcs connecting a point in M to a point in P.
- simplices: a k-simplex is a collection of k isotopy classes of arcs connecting a point in M to a point in P such that one can k representatives pairwise disjoint.

Proposition 5.2. *Let $m \geq 3$, the descending link of a vertex in $\mathscr{SC}(A_{1,m})$ with height $k \geq m$ is isomorphic to $\mathcal{R}(\mathbb{D}, K, M)$ where M is a set of m marked point in the disc $\mathbb{D}$ and K is a set of k punctures.*

Proof. Let $[\Sigma, \phi]$ be a vertex in $\mathscr{SC}(A_{1,m})$ with height $k \geq m$. Without loss of generality, we assume that $\phi = \mathrm{id}$. Observe that Σ has m frontier arcs. Since $k \geq m$ for frontier arcs, $i \in \mathrm{Fr}(\Sigma)$, we give a disk D_i and a puncture p_i such that:

- D_i is a one-punctured disk,
- $D_i \cap \partial\Sigma$ is connected and its intersection with $\mathrm{Fr}(\Sigma)$ equals i,
- the D_i's are pairwise disjoint.

Claim 5.3. Let $I \subset \mathrm{Fr}(\Sigma)$, then there exist an admissible surface Σ_I and ϕ_I an element in $\mathfrak{mod}(A_{1,m})$ such that $[\Sigma, \mathrm{id}] = [\Sigma_I, \phi_I]$ and for each $i \in I$, $\phi_I^{-1}(D_i) = H_i^I$ is an extremal polygon of the rigid structure in Σ_I.

Proof. Observe that on one hand the cardinality of I (denoted $\sharp I$) is $< m$ and on the other hand $k \geq 3$. Furthermore, since we work with $\mathscr{S}(A_{1,m})$, the admissible surface Σ has at least two extremal polygons K_i and K_j (here by extremal polygon we mean that $\Sigma - K$ is connected).

If $\sharp I \leq 2$, then we just send D_i in the K_i where it lies via ϕ_i which is rigid outside H_i. Then we take $\phi_I := \phi_i$ if $I = \{i\}$ and $\phi_I := \phi_i\phi_j$ if $I = \{i, j\}$.

Now assume that $2 < \sharp I < m$. Recall that $\sharp I < k$, then we modify Σ by pushing its polygons contained in distinct strands of $A_{1,m}$ so that we get at least $\sharp I$ extremal polygons.

First observe that the elments in I can be associated bijectively to the set of strands of $A_{1,m}$. Indeed, each frontier arc in I is shared with only one polygon outside Σ and lying in the strand i. We write $I = \{i_1 < ... < i_k\}$, then let χ_I be the map sending successively (from the lower to the greater) the polygon H_i containing the puncture associated to i to an extremal polygon in the strand R_i associated to i. See Figure 42.

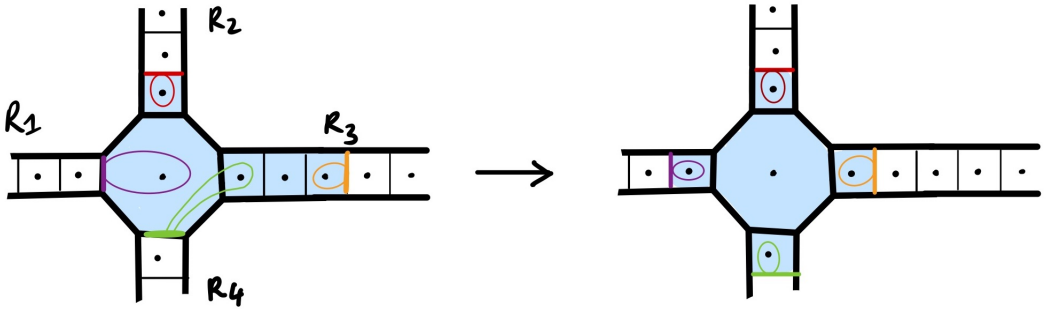


Figure 42

Hence then there exists an admissible surface $\Sigma_I = \chi_I(\Sigma)$ and $\phi_I = \chi_I^{-1}$ such that $[\Sigma, \mathrm{id}] = [\Sigma_I, \phi_I]$ and for each $i \in I$, $\phi_I^{-1}(D_i) = H_i^I$ where H_i^I is an extremal polygon. □

Let $\mathcal{L}$ be the descending link of $[\Sigma, \mathrm{id}]$. By definition, a k-simplex in $\mathcal{L}$ corresponds to k edges containing $[\Sigma, \mathrm{id}]$ and spanning a k-cube together. Let $I \subset \mathrm{Fr}(\Sigma)$, consider the k-simplex S_I in $\mathcal{L}$ spanned by the edges from $[\Sigma, \mathrm{id}] = [\Sigma_I, \phi_I]$ to $[\Sigma_I - H_j^I, \phi_I]$ for j in I. Note that the simplex S_I in $\mathcal{L}$ corresponds to the cube generated by the vertices $[\Sigma_I - \cup_{j\in J} H_j^I, \phi_I]$ where $J \subset I$. We denote by $\mathcal{S}$ the subcomplex of $\mathcal{L}$ generated by the union of all the S_I where $I \subset \mathrm{Fr}(\Sigma)$.

Claim 5.4. The vertices of $\mathcal{S}$ correspond to the vertices $\{[\Sigma_i - H_i^i, \phi_i] | i \in \mathrm{Fr}(\Sigma)\}$.

Proof. Let x be a vertex in $\mathcal{S}$, then x lies in S_I for some $I \subset \mathrm{Fr}(\Sigma)$. By the previous paragraph, x is an edge from $[\Sigma_I, \phi_I]$ to $[\Sigma_I - H_j^I, \phi_I]$ for some j in I. Then observe that Σ_j is sent to Σ by ϕ_j and the previous is mapped to Σ_I by ϕ_I. Moreover, H_j^j is sent to H_j^I by $\phi_I^{-1}\phi_j$ and the latter map is rigid outside Σ_j. Hence, $[\Sigma_j - H_j^j, \phi_j] = [\Sigma_I - H_j^I, \phi_I]$. □

Claim 5.5. The set of simplices $\mathcal{S}$ is a fundamental domain for the action of $\mathrm{Mod}(\Sigma)$ on the set of simplicies of $\mathcal{L}$.

Proof. First take let $I, J \subset \mathrm{Fr}(\Sigma)$, S_I and S_J two simplices of $\mathcal{S}$ in the same $\mathrm{Mod}(\Sigma)$ orbit, i.e. $S_I = gS_J$ for some $g \in \mathrm{Mod}(\Sigma)$. We show that $I = J$ and hence $S_I = S_J$. Let $i \in I$, since $S_I = gS_J$, the vertex in S_I corresponding to the edge from $[\Sigma_I, \phi_I]$ to $[\Sigma_I - H_i^I, \phi_I]$ is sent to the edge from $[\Sigma_J, g\phi_J]$ to $[\Sigma_J - H_j^J, g\phi_J]$ for some $j \in J$. Hence on one hand has $[\Sigma_I - H_i^I, \phi_I] = [\Sigma_J - H_j^J, g\phi_J]$. On the other hand, $(\phi_i)^{-1}g\phi_J$ maps Σ_J to Σ_I and is rigid outside Σ_J. Thus $(\phi_i)^{-1}g\phi_J$ sends H_i^I to H_j^J and $i = j$. So $I \subseteq J$, by symmetry one obtains $I = J$.

Finally, it remains to show that if S is a simplex in $\mathcal{L}$ then it can be written as the translation of a simplex in $\mathcal{S}$ by an element in $\mathrm{Mod}(\Sigma)$. The simplex S corresponds to a cube

in $\mathscr{SC}(A_{1,m})$, let say this cube is generated by the vertices $\{[\Delta \cup \cup_{j\in J} K_j, \psi] | J \subset \{1, ..., q\}\}$ where $[\Sigma, \mathrm{id}] = [\Delta \cup K_1 \cup ... \cup K_q, \psi]$. Let

$$J = \{j \in \mathrm{Fr}(\Sigma) | j \in \mathrm{Fr}(\Sigma) \cap \psi(Ki) \text{ for some } 1 \leq i \leq m\}.$$

For $1 \leq i \leq m$, there exists $j_i \in J$ such that $\psi(K_i)$ and D_{j_i} share the same frontier arcs in Σ. Then, since the D_i are disjoint, there exists an element $\beta \in \mathrm{Mod}(\Sigma)$ that untwists each $\psi(K_i)$ in D_{j_i} for $1 \leq i \leq m$. Then observe that $\phi_I^{-1}\beta\psi$ maps $\Delta \cup K_1 \cup ... \cup K_q$ to Σ_I and K_i to $H^I_{j_i}$. Moreover the latter is rigid outside $\Delta \cup K_1 \cup ... \cup K_q$. Thus:

$$\beta[\Delta \cup K_1 \cup ... \cup K_{i-1} \cup K_{i+1} \cup ... \cup K_q] = [\Sigma_I - H^I_i, \phi_I] \text{ and } \beta S = S_I.$$

□

For each $i \in \mathrm{Fr}(\Sigma)$ we fix a marked point m_i lying in i and α_i an arc from m_i to p_i staying in D_i. Note that one has m marked points and the α_i are pairwise disjoint. One constructs an isomorphism from $\mathcal{L}$ to $\mathcal{R}(\mathbb{D}, K, M)$, where Σ is viewed as a disk, M is the set of marked points and K is the set of punctures. This isomorphism associates to a vertex in $\mathcal{L}$, i.e an edge the form $g($ edge from $[\Sigma, \mathrm{id}]$ to $[\Sigma_i - H^i_i, \phi_i])$, the element $g\alpha_i$ in $\mathcal{R}(\mathbb{D}, K, M)$. It is indeed bijective because of claims 5.5, 5.4 and also because the set $\{\alpha_i | i \in \mathbb{Z}/m\mathbb{Z}\}$ is a strict fundamental domain for the action of $\mathrm{Mod}(\mathbb{D})$ on $\mathcal{R}(\mathbb{D}, K, M)$. Moreover, it is well defined because the stabilizers of the action of $\mathrm{Mod}(\Sigma)$ (resp. $\mathrm{Mod}(\mathbb{D})$) on $\mathcal{L}$ (resp. $\mathcal{R}(\mathbb{D}, K, M)$) coincides. Indeed the stabilizers of the simplices S_I, resp. $\{\alpha_i | i \in I\}$, are $\mathrm{Mod}(\Sigma - \cup_{i\in I} D_i)$, resp. $\mathrm{Mod}(\mathbb{D} - \cup_{i\in I} N(\alpha_i))$, where the $N(\alpha_i)$ is a tubular neighbourhood of α_i so they can be viewed as the D_i. To check that the stabilizer of S_I is indeed $\mathrm{Mod}(\Sigma - \cup_{i\in I} D_i)$, observe that if $[\Sigma_I - H^I_i, g\phi_I] = [\Sigma_I - H^I_i, \phi_I]$ then $\phi_I^{-1} g \phi_I$ maps $\Sigma_I - H^I_i$ to itself. Since ϕ_I sent H^I_i to D_i it follows that g must stabilizes D_i. □

Proposition 5.6. *Let $k \geq 3$, M be a set of m marked point in the disc $\mathbb{D}$ and K a set of k punctures. Then $\mathcal{R}(\mathbb{D}, K, M)$ is simply connected.*

Proof. Let $n \in M$ and $(k_1, ..., k_k)$ be an enumeration of the k points in K. Let $\mathcal{R}_0$ be the subcomplex of $\mathcal{R}(\mathbb{D}, K, M)$ generated by the vertices represented by arcs connecting a point in $M - n$ to a point in $K - k_1$. We proceed by induction on m the cardinality of M. If $\sharp M = 1$, then $\mathcal{R}(\mathbb{D}, K, M)$ is trivial. Now assume that $\sharp M \geq 2$.

Claim 5.7. It is enough to show that $\mathcal{R}_0$ is simply connected

Proof. Let $\mathcal{R}_1$ be the subcomplex of $\mathcal{R}(\mathbb{D}, K, M)$ generated by $\mathcal{R}_0$ and the vertices represented by arcs connecting n to k_1. By definition, the latter vertices are not adjacent. Thus, we obtain $\mathcal{R}_1$ from $\mathcal{R}_0$ by gluing cones over the links of the vertices represented by arcs connecting n to k_1. Now this links are isomorphic to $\mathcal{R}(\mathbb{D} \cup \{k_1\}, K - \{k_1\}, M - \{n\})$ which is simply connected by the induction hypothesis.

Then let $\mathcal{R}_2$ be the subcomplex generated by $\mathcal{R}_1$ and the vertices represented by arcs connecting n to a point in $K - \{k_1\}$. By definition, the latter vertices are not adjacent. Thus, we obtain $\mathcal{R}_2$ from $\mathcal{R}_1$ by gluing cones over the links of the vertices represented by arcs connecting n to $K - \{k_1\}$. Now this links are isomorphic to $\mathcal{R}(\mathbb{D} \cup \{k\}, K - \{k_1, k\}, M - \{n\})$ where $k \in K - \{k_1\}$ which is simply connected by the induction hypothesis.

Finally, let $\mathcal{R}_3$ be the subcomplex generated by $\mathcal{R}_2$ and the vertices represented by arcs connecting $M-\{n\}$ to k_1. By definition, the latter vertices are not adjacent. Thus, we obtain $\mathcal{R}_3$ from $\mathcal{R}_2$ by gluing cones over the links of the vertices represented by arcs connecting a point in $M-\{n\}$ to k_1. Now this links are isomorphic to $\mathcal{R}(\mathbb{D}\cup\{k_1\}, K-\{k_1\}, M-\{n'\})$, with $n'\in M-\{n\}$, the latter is simply connected by the induction hypothesis.

Now, observe that $\mathcal{R}_3$ is the whole complex $\mathcal{R}(\mathbb{D}, K, M)$. Since we build $\mathcal{R}_3$ by successively gluing cones over $\mathcal{R}_0$ and then over simply connected spaces we deduce that to show that $\mathcal{R}(\mathbb{D}, K, M)$ is simply connected it remains to show that $\mathcal{R}_0$ is simply connected. □

Take a map $\psi\colon \mathbb{S}^1 \longrightarrow \mathcal{R}_0$. We want to show that ψ is homotopically trivial. By the simplicial approximation theorem, one can assume that ψ is simplicial. Moreover by Lemma 5.15 and Claim 5.25 in [GLU22] it is injective on each simplex. Let γ be an arc from n to k_1. Observe that the star of γ, denoted $\mathrm{star}(\gamma)$ is contractible. We want to show that the image of ψ lies in $\mathrm{star}(\gamma)$.

If all the representatives of the vertices in $\operatorname{Im}\psi$ have no intersection with γ we are done. Indeed, the representatives are disjoint so they give a simplex which has γ as a face. Thus, in this case, $\operatorname{Im}\psi$ lies in $\mathrm{star}(\gamma)$. Now assume that some representatives of the vertices in $\operatorname{Im}\psi$ intersect γ. Let $x\in\mathbb{S}^1$ such that $\psi(x)$ intersects γ closest to k_1. Note that the arcs representing an element of $\operatorname{Im}\psi$ have their endpoints distinct from n and k_1 by definition of $\mathcal{R}_0$. Denote by α the arc from m' to k_2 represented by the vertex $\psi(x)$. Now, take a small disc in S around p such that

- it is disjoint from all the arcs representing an element in $\operatorname{Im}\psi$ that is disjoint from the presentative α of $[\alpha]=\psi(x)$,
- its boundary intersects α in a sub-arc β.

Let α' be the arc obtained by replacing the sub-arc β in α by its complement in the boundary of the disc. See Figure 43. Since ψ is simplicial and injective on simplices, $\psi(\mathrm{Lk}(x))$ lies in

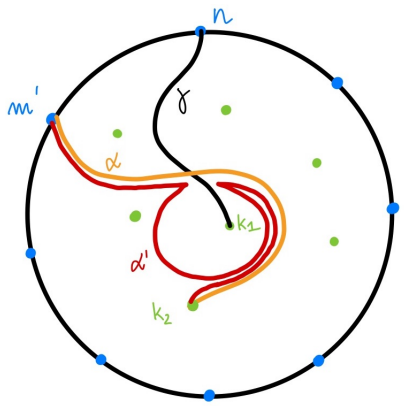


Figure 43

$\mathrm{Lk}(\psi(x))=\mathrm{Lk}([\alpha])$ and by construction of α' one has $\psi(\mathrm{Lk}(x))\subseteq\mathrm{Lk}([\alpha'])$. Now we define

$$\begin{aligned}\psi'\colon \mathbb{S}^1 &\longrightarrow \mathcal{R}_0\\ z &\longmapsto \begin{cases}\psi(z) & \text{if } z\neq x,\\ \text{vertex represented by } [\alpha'] & \text{if } z=x.\end{cases}\end{aligned}$$

Claim 5.8. The two maps ψ and ψ' are homotopy equivalent in $\mathcal{R}(\mathbb{D}, K, M)$.

Proof. The intersection L of $\mathrm{Lk}([\alpha])$ and $\mathrm{Lk}([\alpha'])$ is isomorphic to $\mathcal{R}(\mathbb{D}\cup\{n, m'\}, M-\{n, m'\}, K-\{k_1, k_2\})$ which is simply-connected by the induction hypothesis. Thus, the common restriction of ψ and ψ', i.e. $\psi : \mathrm{Lk}(x) \longrightarrow L$ and $\psi' : \mathrm{Lk}(x) \longrightarrow L$ is homotopically trivial. Hence there exists $\phi\colon \mathrm{star}(x) \longrightarrow L$ such that $\phi(y) = \psi(y) = \psi'(y)$ for all $y \in \mathrm{Lk}(x)$. Now, $\mathrm{Im}\,\phi$ lies in L so in the link of $\psi(x)$. Moreover $\mathrm{star}(\psi(x))$ is contractible so the restriction $\psi\colon \mathrm{star}(x) \longrightarrow \mathrm{star}(\psi(x))$ is equal to ϕ up to a homotopy that fixes ψ outside $\mathrm{star}(x)$. Similarly, $\mathrm{Im}\,\phi$ lies in L so in the link of $\psi'(x)$. Moreover $\mathrm{star}(\psi'(x))$ is contractible so the restriction $\psi'\colon \mathrm{star}(x) \longrightarrow \mathrm{star}(\psi'(x))$ is equal to ϕ up to an homotopy that fixes ψ' outside $\mathrm{star}(x)$. Thus ψ and ψ' are homotopy equivalent. □

By taking ψ' we decrease the number of intersection points between the representatives of the elements in the image of ψ' and γ. So by iterating the process we obtain a map from $\mathbb{S}$ to $\mathcal{R}_0$ with an image lying in $\mathrm{star}(\gamma)$ and which is homotopy equivalent to ψ. □

Definition 5.9. Let $\mathscr{SC}^k(A_{1,m})$ be the cube complex generated by the vertices whose height is less than or equal to k. Let k equal to m, then $\mathscr{SC}^m(A_{1,m})$ is called *the height bounded* complex.

Proposition 5.10. *Let $m \geq 3$, the cube complex $\mathscr{SC}^m(A_{1,m})$ is simply connected.*

Lemma 5.11. *Let X be a CW-complex and $\tilde{X}$ obtained from X by gluing finitely many cones over simply connected sub-complexes of X. Then, any loop in X which is contractible in $\tilde{X}$ is already contractible in X. In other words, this means that any commutative square*

$$\begin{array}{ccc} \mathbb{S}^1 & \xrightarrow{\psi} & X \\ \downarrow & \nearrow & \downarrow \\ \mathbb{D}^2 & \xrightarrow[\tilde{\psi}]{} & \tilde{X} \end{array}$$

can be filled with a dashed arrow, making the lower triangle commute and the upper diagram commute up to homotopy.

Proof of the Lemma. It suffices to consider the case where $\tilde{X}$ is obtained from X by gluing a single cone. The general case follows by induction.

Thus, assume that $\tilde{X}$ is obtained from X by gluing a cone over a simply connected sub-complex $Y \subset X$. Applying Seifert-Van Kampen theorem to $\tilde{X} = X \cup CY$ where CY is the glued cone over Y and $X \cap CY = Y$, we get

$$\pi_1(\tilde{X}) \cong \pi_1(X) \underset{\pi_1(Y)}{*} \pi_1(CY) \cong \pi_1(X)$$

since Y and CY are contractible, and the isomorphism $\pi_1(X) \cong \pi_1(\tilde{X})$ is induced by the inclusion $X \hookrightarrow \tilde{X}$. Thus, a loop in X which is trivial in $\pi_1(\tilde{X})$ must already be trivial in $\pi_1(X)$, which concludes the proof. □

Proof of the Proposition. Let $m \geq 3$ and take a map $\psi : \mathbb{S}^1 \longrightarrow \mathscr{SC}^m(A_{1,m})$. We want to show that it is homotopically trivial. By the theorem of cellular approximation, we may assume that the image of ψ lies in the underlying graph of $\mathscr{SC}^m(A_{1,m})$. Now, by Proposition 5.2 and Theorem 3.3 in [GLU22] we know that $\mathscr{SC}(A_{1,m})$ is contractible. Hence there exists a map $\tilde{\psi}\colon \mathbb{D}^2 \longrightarrow \mathscr{SC}(A_{1,m})$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathbb{S}^1 & \xrightarrow{\psi} & \mathscr{SC}^m(A_{1,m}) \\ \downarrow & & \downarrow \\ \mathbb{D}^2 & \xrightarrow[\tilde{\psi}]{} & \mathscr{SC}(A_{1,m}) \end{array}$$

Again by simplicial approximation theorem, we take $\tilde{\psi}$ simplicial. If $\mathrm{Im}(\tilde{\psi})$ lies in $\mathscr{SC}^m(A_{1,m})$ we are done.

Otherwise, let $n > m$ be the maximal height of a vertex in $\mathrm{Im}(\tilde{\psi})$ (which exists since $\mathbb{D}^2$ is compact). Thus $\mathrm{Im}(\tilde{\psi})$ lies in $\mathscr{SC}^n(A_{1,m})$. By Lemma 2.5 in [BB97], $\mathscr{SC}^{i+1}(A_{1,m})$ is obtained from $\mathscr{SC}^i(A_{1,m})$ by gluing cones over the descending links of vertices of height $i+1$ for $i \geq 1$. Since $\mathrm{Im}(\tilde{\psi})$ is compact, it meets only finitely many of those cones.

Now, by Proposition 5.2 the descending link of v a vertex with height $\geq m$, labelled $\mathrm{Lk}_{\downarrow}(v)$, is isomorphic to $\mathcal{R}(\mathbb{D}, K, M)$ where M is a set of m marked point of the discs and K a set of n punctures. By Proposition 5.6, the latter is simply connected thus so is $\mathrm{Lk}_{\downarrow}(v)$.

By Lemma 5.11, $\tilde{\psi}$ is homotopic to a map $\tilde{\psi} : \mathbb{D}^2 \longrightarrow \mathscr{SC}^{n-1}(A_{1,m})$, such that $\tilde{\psi} \mid_{\mathbb{S}^1} = \psi$.

By proceeding by induction (until the vertices of height $m+1$, in order for Proposition 5.2 to hold) we obtain $\tilde{\psi} : \mathbb{D}^2 \longrightarrow \mathscr{SC}^m(A_{1,m})$. □

5.2 Computing the presentation for the case $m = 3$

In [Bro84], K.S Brown gives a method to compute a presentation of a group that acts on a (simply)-connected complex. In the previous paragraph we have shown that the height bounded subcomplex of $\mathscr{SC}(A_{1,m})$ is simply connected. It is a good idea to use it to compute the presentation of $\mathfrak{mod}(A_{1,m})$ with the algorithm of Brown. Indeed, because the height of the vertices is bounded, the algorithm becomes relatively easy to compute. Here we treat the case $m = 3$.

Let M be the central polygon, R_i the ith strand with $i = 1, 2, 3$. Consider the situation in Figure 44.

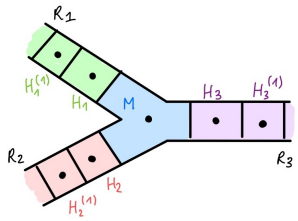


Figure 44

Example 5.12. Let $\beta \in \mathfrak{mod}(A_{1,m})$ be the element described in Figure 45. Let α_i be the element in $\mathfrak{mod}(A_{1,m})$ which translate by one unit the punctures of the strings $R_i \cup R_{i+1}$in the counterclockwise direction. The element α_1 is represented in Figure 46. △



Figure 45

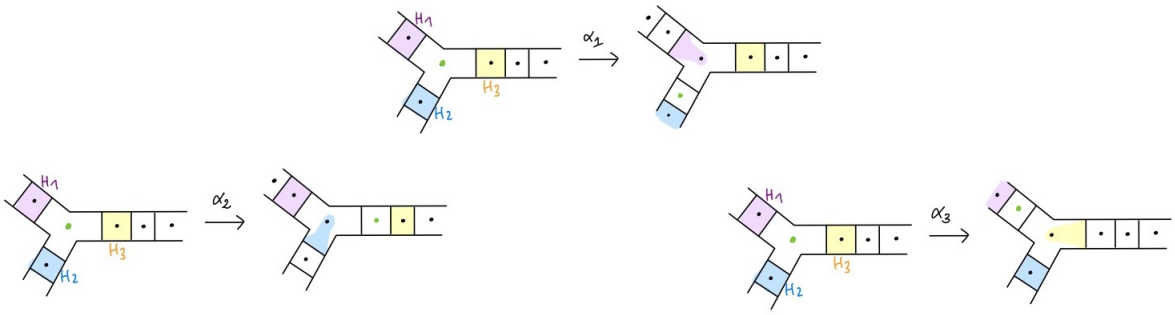


Figure 46

Theorem 5.13 (Theorem 1 from [Bro84])**.** *Let G be a group acting on a simply-connected CW complex G so that the edges of X are not inverted by the action. We define the following objects:*

- *Let T be a tree whose set of vertices is V a set of representatives of the vertices of X modulo G.*
- *Let E be a set of edges that form a set of representatives of $X \mod G$ such that each starting point of $e \in E$ lies in T.*
- *For each $e \in E$ we fix an element $g_e \in G$ such that $t(e) = g_e w(e)$ where $t(e)$ is the ending point of e and $w(e)$ is its representative in T.*
- *Let F be a representative of the 2-cells of X modulo G. To each τ we associate an element r_τ by building it from the elements g_e associated to the edges of τ. See a description of the method in [Bro84].*

Then, G is generated by the isotropy subgroups G_v $(v \in V)$ and the elements g_e $(e \in E)$ subject to the following relations:

1. *$g_e = 1$ if e is an edge of T.*
2. *$g_e^{-1} i_e(h) g_e = c_e(h)$ for all h in the stabilizer G_e, where $e \in E$, i_e is the inclusion of G_e into $G_{o(e)}$ and c_e is the inclusion of G_e into $G_{w(e)}$.*
3. *$r_\tau = 1$ for any τ in F.*

Theorem 5.14. *The group $\mathfrak{mod}(A_{1,3})$ admits a finite presentation with generators α_1, β, γ, t, σ_1, σ_2 and the following relations:*

1. *$\beta^3 = 1$, $t^3 = 1$*

2. $\sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2$

3. $t\sigma_1 t^{-1} = \sigma_1\sigma_2\sigma_1$

4. $t\sigma_2 t^{-1} = \sigma_1$

5. $\alpha_1\sigma_1\alpha_1^{-1} = \sigma_1$.

6. $\sigma_1\gamma = \gamma\sigma_1$

7. $\sigma_1^2 = \gamma^3$

To prove Theorem 5.14 we need some technical lemmata.

Lemma 5.15. *Two vertices in $\mathscr{SC}^3(A_{1,3})$ are in the same $\mathfrak{mod}(A_{1,3})$-orbit if and only if they have the same height.*

Proof. Let $[\Sigma, \phi]$ and $[\Gamma, \pi]$ be two vertices in $\mathscr{SC}^3(A_{1,3})$ in the same $\mathfrak{mod}(A_{1,3})$-orbit. Then, there exists $g \in \mathfrak{mod}(A_{1,3})$ such that $[\Sigma, \phi] = g[\Gamma, \pi]$. So $\phi^{-1}g\pi$ maps Γ to Σ. Consequently Γ and Σ have the same number of punctures. Conversely, assume that $[\Sigma, \phi]$ and $[\Gamma, \pi]$ have the same height. Since we work with the spine of $A_{1,3}$, Σ and Γ have the same number of punctures, the same number of boundary components and so the same euler characteristic. Moreover all connected components the complement are homeomorphic. By the theorem of classification of surfaces, there exists $g \in \mathfrak{mod}(A_{1,3})$ such that $g\Sigma = \Gamma$ so $[\Sigma, \mathrm{id}] = g[\Gamma, \mathrm{id}]$. Consequently, $[\Sigma, \phi] = \phi[\Sigma, \mathrm{id}]$ and $[\Gamma, \pi] = \pi[\Gamma, \phi]$ are in the same $\mathfrak{mod}(A_{1,3})$-orbit. □

Let E be the set of all edges of $\mathscr{SC}^3(A_{1,3})$ modulo the action by $\mathfrak{mod}(A_{1,3})$. Consider in Figure 47 the following tree T in $\mathscr{SC}^3(A_{1,3})$ whose underlying set of vertices is a set of representatives for $\mathscr{SC}^3(A_{1,3})$ modulo the action by $\mathfrak{mod}(A_{1,3})$.

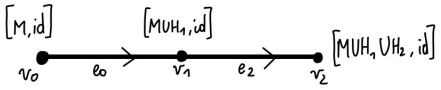


Figure 47

Lemma 5.16. *All the edges of $\mathscr{SC}^3(A_{1,3})$ starting from a vertex of height one are in the same $\mathfrak{mod}(A_{1,3})$-orbit as e_0.*

Proof. Consider an edge e in $\mathscr{SC}^3(A_{1,3})$ starting with a vertex of height one. Necessarily e is an edge from $[M, \phi]$ to $[M \cup K, \phi]$ where $K = H_j$ for some $j = 1, 2, 3$, and $\phi \in \mathfrak{mod}(A_{1,3})$. Let $\operatorname{supp}\phi$ be the support of ϕ.

- If $\operatorname{supp}\phi \cap (M \cup K) = \emptyset$: since ϕ is rigid outside $\operatorname{supp}\phi$, it sends M to M and $M \cup K$ to $M \cup H_k$ for some k. Then there is $l = 1, 2, 3$ such that $\beta^l(M \cup H_k) = M \cup H_1$ where β is as in Example 5.12. Since $\beta^l(M) = M$, one has $\beta^l(e) = e_0$.

- If $\operatorname{supp}\phi \cap (M \cup K) \neq \emptyset$: then either one of the following statements holds

  1. $M \cup K \subseteq \operatorname{supp}\phi$,

2. $M \subseteq \operatorname{supp}\phi$ and $K \cap \operatorname{supp}\phi = \emptyset$,
3. $K \subseteq \operatorname{supp}\phi$ and $M \cap \operatorname{supp}\phi = \emptyset$.

Assume that 1 holds. If $\operatorname{supp}\phi = M \cup K$ we are done. Otherwise, since ϕ is rigid outside $\operatorname{supp}\phi$ the latter contains M. Then one can untwist $\phi(M)$ in M via some τ in $\mathfrak{mod}(A_{1,3})$. Then $\tau(\phi(K))$ is adjacent to M and has the same euler characteristic than the H_j. Thus there is τ' that untwists $\tau\phi(K)$ in a polygon adjacent to M and leaves M fixed. Thus $\beta^l\tau'\tau(e) = e_0$ for some $l = 1, 2, 3$.

Then, assume that 2 is true. Since ϕ is rigid outside $\operatorname{supp}\phi$, $\phi(K) = H_j$ is a polygon adjacent to $\phi(M)$. If $\phi(M) = M$ we proceed as previously. Otherwise, $\phi(M) \neq M$ and $M \subseteq \operatorname{supp}\phi$ so $\operatorname{supp}\phi \geq 2$. By theorem of classification of surface, one can untwist $\phi(M)$ in M via $\tau \in \mathfrak{mod}(A_{1,3})$ which is rigid outside $\operatorname{supp}\phi$ so $\tau(\phi(M)) = M$ and $\tau(\phi(K)) = H_j$. Hence, $\beta^l\tau(e) = e_0$.

Finally, assume that 3 holds. Since ϕ is rigid outside $\operatorname{supp}\phi$, $\phi(M) = M$ and $\phi(K)$ is adjacent to M. If $\phi(K) = K$ we are done, otherwise $\operatorname{supp}\phi \geq 2$ and then by Lemma 3.20 there exists χ such that $[\operatorname{supp}\phi \cup M, \chi] = [\Omega \cup M, \mathrm{id}]$, $\chi(M) = M$ and $\chi(K)$ is an extremal polygon, i.e. $\chi(\phi(K)) = H_j$ (still adjacent to M). So $\chi(e) = e_0$. □

Lemma 5.17. *A set of representatives for the set of edges in $\mathscr{SC}^3(A_{1,3})$ starting with a vertex of height 2 modulo the action by $\mathfrak{mod}(A_{1,3})$ is given by $\{e_1, e_2\}$. Where e_2 is represented in Figure 47 and e_1 is the edge from $[M \cup H_1, \mathrm{id}]$ to $[M \cup H_1 \cup H_1^{(1)}, \mathrm{id}]$.*

Proof. Take an edge e in $\mathscr{SC}^3(A_{1,3})$, with starting vertex, called $o(e)$, with height 2. Then e is an edge from $[M \cup H_j, \phi]$ to $[M \cup H_j \cup K, \phi]$ where K either is H_k for $k \neq j$ or $K = H_j^1$, where H_j^1 is the polygon adjacent to H_j in the jth strand of $A_{1,3}$. If K equals $K = H_j^1$ then $e = \phi\beta^l(e_2)$ for some $l = 1, 2, 3$. If K equals H_k for $k \neq j$ then again one can find a power of β such that $e = \phi\beta^l(e_2)$. Then, e_1 and e_2 are not in the same G-orbit because we need to preserve the adjacency. □

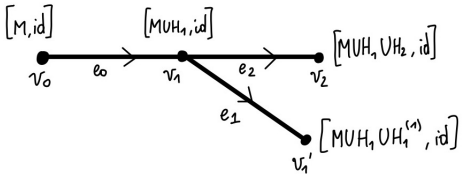


Figure 48

By Lemma 5.16 and Lemma 5.17, E is represented in Figure 48. Now, for each $e \in E$ we pick an element $g_e \in \mathfrak{mod}(A_{1,3})$ such $t(e) = g_e w_e$ where $t(e)$ designes the terminal vertex of e and w_e is a vertex of T.

$$g_e = \begin{cases} \mathrm{id} & \text{if } e \in T \\ \alpha_1^{-1} & \text{if } e = e_1 \end{cases}$$

Let F be a set of representatives for the 2-cells of $\mathscr{SC}^3(A_{1,3})$ modulo the action by $\mathfrak{mod}(A_{1,3})$.

Lemma 5.18. *There is a single $\mathfrak{mod}(A_{1,3})$-orbit for the 2-cells of $\mathscr{SC}^3(A_{1,3})$ modulo the action by $\mathfrak{mod}(A_{1,3})$. Then we choose Figure 49 as a representative.*

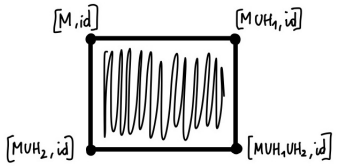


Figure 49

Proof. The proof is very similar to the one of the preceding lemma. □

Proof of Theorem 5.14. We follow the method given Theorem 5.13, with the action of $G := \mathfrak{mod}(A_{1,3})$ on $\mathscr{SC}^3(A_{1,3})$. By Proposition 5.10, X is simply connected. We recall that we study $\mathscr{SC}^3(A_{1,3})$ which is height-oriented. We apply Theorem 5.13 to find the generators and the relators.

- Generators By Lemma 5.15, Lemma 5.16 and Lemma 5.17 the generators are given by the isotropy subroups of v_0, v_1 and v_2 represented in Figure 47 and the elements g_e with $e \in E$. Recall that $g_{e_1} = \alpha_1^{-1}$ Thus it remains to compute the isotropy groups.

  $G_{v_0} = \{g \in \mathfrak{mod}(A_{1,3}) \mid gM = M \text{ and g is rigid outside } M\} = \{\text{rigid homeomorphisms }\}$
  $= \{\beta, \beta^2, \mathrm{id}\}$

  Claim 5.19. There is an exact sequence

  $$1 \longrightarrow \mathcal{B}_2 \xrightarrow{i} G_{v_1} \xrightarrow{\phi} \mathcal{A}_3 \longrightarrow 1$$

  where $\mathcal{A}_3$ is the group of cyclic permutations.

  Proof. Let ϕ be the surjective morphism which associates to each element $g \in G_{v_1}$ its underlying permutation on the strands of $A_{1,3}$ at infinity. Recall that the elements of G_{v_1} are the ones preserving $M \cup H_1$ and that are rigid outside $M \cup H_1$. Then the kernel of ϕ are the braided elements permuting the two punctures of $M \cup H_1$ and being the identity outside.

  □

  Let H be the group with the following presentation:

  $$\left\langle a, b \;\middle|\; \begin{array}{c} ab = ba \\ a^2 = (b^3)^{-1} \end{array} \right\rangle.$$

  Claim 5.20. The isotropy group G_{v_1} is isomorphic to H.

  Proof. We define the following homomorphism,

  $$\begin{aligned} \pi \colon H &\longrightarrow G_{v_1} \\ a &\longmapsto \sigma := \alpha_2\alpha_3\alpha_2^{-1}\alpha_3^{-1} \\ b &\longmapsto \gamma := (\alpha_3\alpha_2\beta). \end{aligned}$$

The element σ is given in Figure 50. Observe that σ twists the two punctures of $M \cup H_1$. It is the image in G_{v_1} of the generator of the braid group $\mathcal{B}_2$ from the short exact sequence of Claim 5.19. The element γ is given in Figure 51. Clearly, its image under ϕ in the short exact sequence of Claim 5.19 generates $\mathcal{A}_3$. Hence, those elements generates G_{v_1} since they generate the quotient $\mathcal{A}_3$ and the kernel $\mathcal{B}_2$.

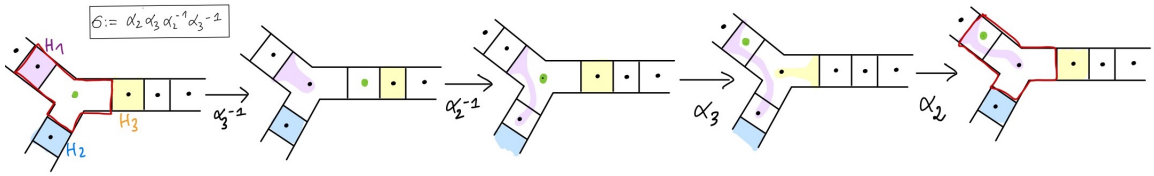


Figure 50

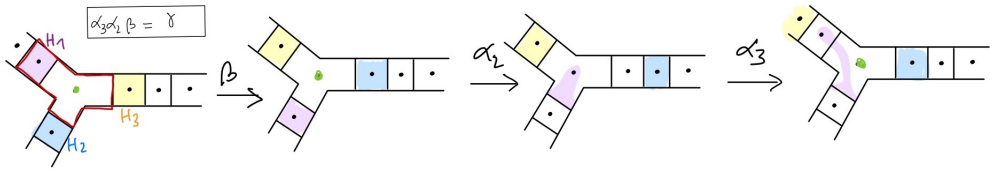


Figure 51

Now it remains to show that σ and γ satisfy the same relations as those from a and b and that we do not have any other relation.

That σ and γ commute is shown in Figure 52. The relation $\sigma^2 = \gamma^3$ is shown in Figure

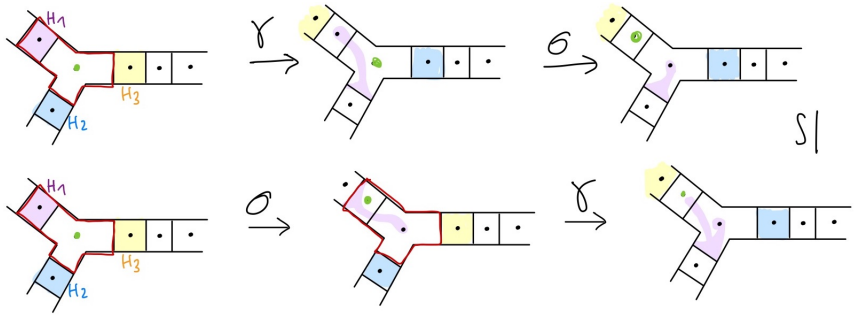


Figure 52

53.

Now, if σ and γ satisfy another relation, the latter must be of the form $\sigma^n \gamma^m = 1$ since the two elements commute. Then

$$\phi(\sigma^n \gamma^m) = \phi(\gamma^m) = 1,$$

since σ^n lies in the kernel of ϕ. Since $\phi(\gamma)$ generates $\mathcal{A}_3$, this implies that m is a multiple of 3. Thus we can replace γ^m by some power of σ (using $\sigma^2 = (\gamma^3)^{-1}$) and the relation becomes $\sigma^k = 1$ for some k. Since $\sigma \in \mathcal{B}_2 = \ker(\phi)$ and braid groups have no torsion, we obtain $k = 0$. Therefore, there is no other relation.

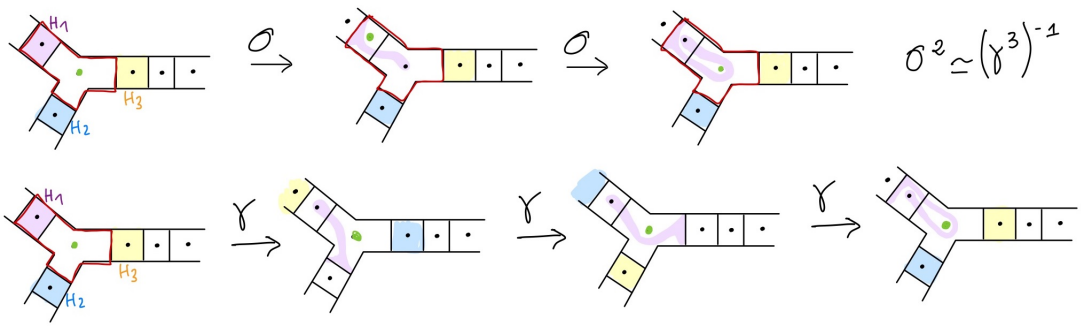


Figure 53

□

Hence, we conclude from Claims 5.19 and 5.20 that G_{v_1} has the following presentation:

$$\left\langle \sigma, \gamma \;\middle|\; \begin{array}{c} \sigma\gamma = \gamma\sigma \\ \sigma^2 = (\gamma^3)^{-1} \end{array} \right\rangle.$$

Claim 5.21. There is an exact sequence

$$1 \longrightarrow \mathcal{B}_3 \xrightarrow{i} G_{v_2} \xrightarrow{\phi} \mathcal{A}_3 \longrightarrow 1$$

where $\mathcal{A}_3$ is the group of cyclic permutations. There is a section $s : \mathcal{A}_3 \longrightarrow G_{v_2}$.

Proof. Let ϕ be the surjective morphism which associates to each element $g \in G_{v_2}$ its underlying permutation on the strands of $A_{1,3}$ at infinity. Recall that the elements of G_{v_2} are the ones preserving $M \cup H_1 \cup H_2$ and being rigid outside $M \cup H_1 \cup H_2$. Then the kernel of ϕ are the braided elements permuting the three punctures of $M \cup H_1 \cup H_2$. Then the morphism:

$$\begin{aligned} s \colon \mathcal{A}_3 &\longrightarrow G_{v_2} \\ t &\longmapsto \alpha_3\beta \end{aligned}$$

gives a section for our exact sequence. The element $(\alpha_3\beta)^3$ is isotopic to the identity see Figure 54. □

Hence we deduce from the previous claim that $G_{v_2} \cong \mathcal{B}_3 \rtimes \mathbb{Z}/3\mathbb{Z}$. The generators of G_{v_2} are given by σ_1, σ_2, t with respect to the following relations:

- $t^3 = 1$
- $\sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2$
- $t\sigma_1 t^{-1} = i^{-1}(s(t)i(\sigma_1)s(t^{-1}))$
- $t\sigma_2 t^{-1} = i^{-1}(s(t)i(\sigma_2)s(t^{-1}))$

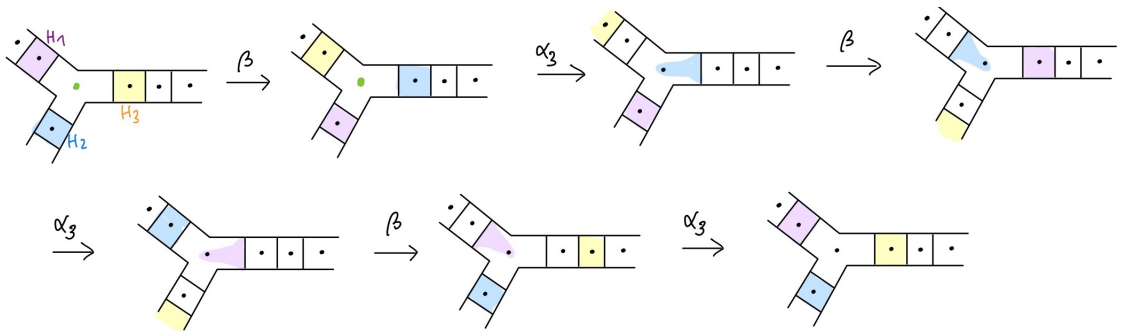


Figure 54

where $s(t) = \alpha_3\beta$, $i(\sigma_1)$ is the braided element twisting the two punctures of $M \cup H_1$, i.e. $\alpha_2\alpha_3\alpha_2^{-1}\alpha_3^{-1}$, and $i(\sigma_2)$ is the braided element twisting the two punctures of $M \cup H_2$, i.e $\alpha_3\alpha_1\alpha_3^{-1}\alpha_1^{-1}$.

Finally, observe that the relations in the presentation of G_{v_2} are:

- $t\sigma_1 t^{-1} = i^{-1}(s(t)i(\sigma_1)s(t^{-1}) = i^{-1}(\alpha_3\beta\sigma(\alpha_3\beta)^2) = 1$,
- $t\sigma_2 t^{-1} = i^{-1}(s(t)i(\sigma_2)s(t^{-1})) = i^{-1}(\alpha_3\beta\ \alpha_3\alpha_1\alpha_3^{-1}\alpha_1^{-1}\ (\alpha_3\beta)^2) = \sigma_1$.

See Figures 55 and 56 for the details.

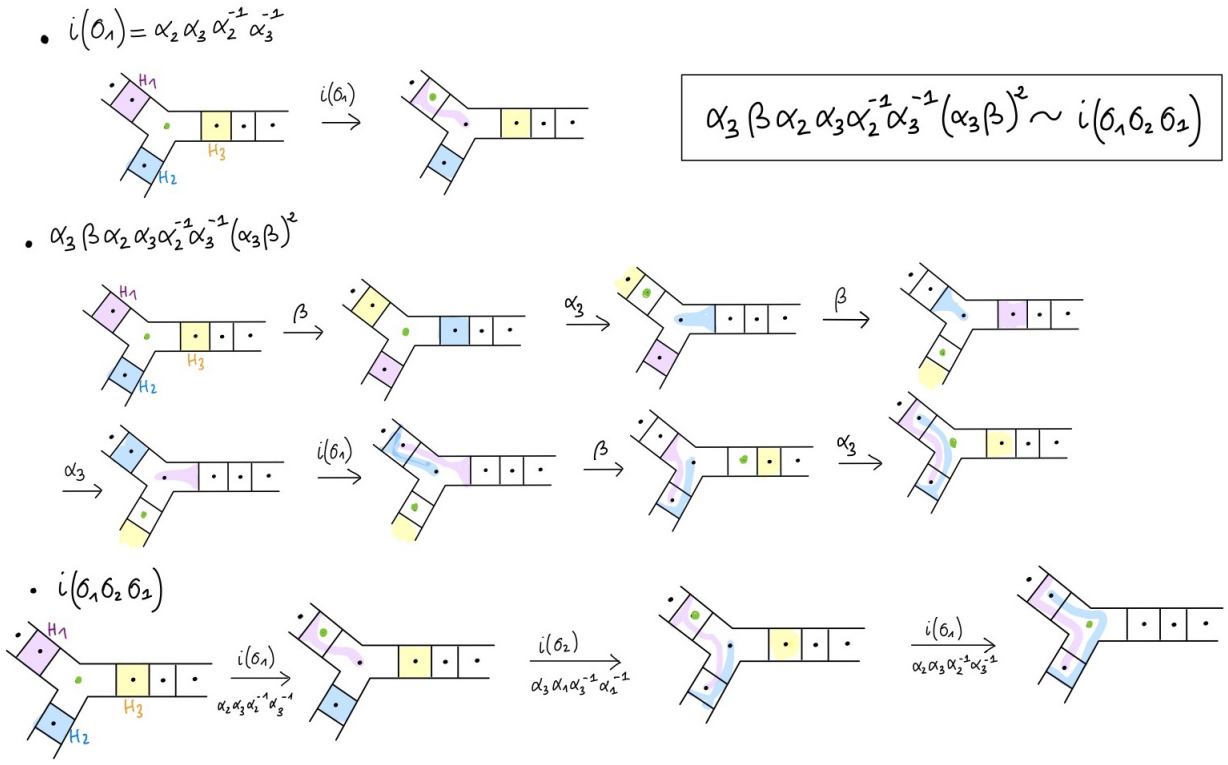


Figure 55

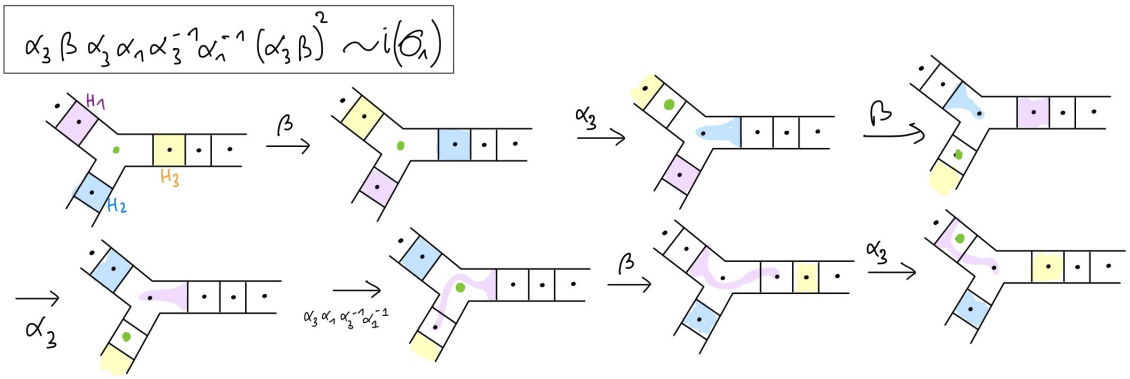


Figure 56

- Relators: By Theorem 5.13 the relators are constructed from the relations $g_e^{-1} i_e(h) g_e = c_e(h)$ for all h in the stabilizer G_e, where $e \in E$, i_e is the inclusion of G_e into $G_{o(e)}$ and c_e is the inclusion of G_e into $G_{w(e)}$. Moreover, we also obtain a relator r_τ from each representative τ of the 2-cells.

  To obtain the relations induced from the stabilizers of the edges in E, by Lemma 5.16 and Lemma 5.17, we need to compute G_{e_0}, G_{e_1} and G_{e_2}:

$$\begin{aligned} G_{e_0} &= \{g \in \mathfrak{mod}(A_{1,3}) \mid gM = M,\ g([M \cup H_1, \mathrm{id}]) = [M \cup H_1, \mathrm{id}], \text{ g rigid outside } M\} \\ &= G_{v_0} \cap G_{v_1} = \{\mathrm{id}\}, \end{aligned}$$

$$\begin{aligned} G_{e_1} &= \{g \in \mathfrak{mod}(A_{1,3}) \mid g([M \cup H_1, \mathrm{id}]) = [M \cup H_1, \mathrm{id}],\ \text{g is rigid outside } M \cup H_1 \text{ and} \\ &\qquad g([M \cup H_1 \cup H_1', \mathrm{id}]) = [M \cup H_1 \cup H_1', \mathrm{id}]\} \\ &= \{\text{elements } g \in \mathfrak{mod}(A_{1,3}) \text{ braiding } M \cup H_1 \text{ and being identity outside}\} \\ &\cong \mathcal{B}_2, \end{aligned}$$

$$\begin{aligned} G_{e_2} &= \{g \in \mathfrak{mod}(A_{1,3}) \mid g([M \cup H_1, \mathrm{id}]) = [M \cup H_1, \mathrm{id}],\ \text{g is rigid outside } M \cup H_1 \text{ and} \\ &\qquad g([M \cup H_1 \cup H_2, \mathrm{id}]) = [M \cup H_1 \cup H_2, \mathrm{id}]\} \\ &= \{\text{elements } g \in \mathfrak{mod}(A_{1,3}) \text{ braiding } M \cup H_1 \text{ and being identity outside}\} \\ &\cong \mathcal{B}_2. \end{aligned}$$

  Thus, this gives two relations:

$$\alpha_1 \sigma \alpha_1^{-1} = g_{e_1}^{-1} i_{e_1}(\sigma) g_{e_1} = c_{e_1}(\sigma) = \sigma$$

$$\sigma = \mathrm{id}\,\sigma\,\mathrm{id} = g_{e_2}^{-1} i_{e_2}(\sigma') g_{e_2} = c_{e_2}(\sigma') = \sigma_1$$

  Finally, to obtain the remaining relations we study the representatives of the 2-cells. By Lemma 5.18 one has only one representative τ in Figure 49. The associated r_τ is a word $\bar{g_\tau}$ in term of generators and an element $g_\tau \in \mathfrak{mod}(A_{1,3})$.

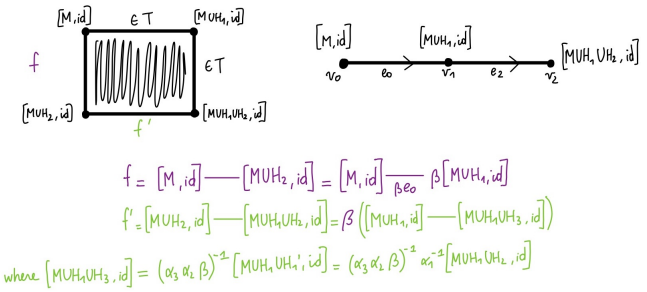


Figure 57

From the description of Theorem 5.13, we deduce that $g_\tau = \beta(\alpha_3\alpha_2\beta)^{-1}\alpha_1^{-1}$, see Figure 57 so $\bar{g_\tau} = \beta(t')^{-1}\alpha_1^{-1}$. Then the relation $r_\tau := \bar{g_\tau}g_\tau^{-1} = 1$ is equivalent to $\beta\gamma\alpha_1^{-1} = \bar{g_\tau} = g_\tau = \beta(\alpha_3\alpha_2\beta)^{-1}\alpha_1^{-1}$.

□

5.3 Finite presentability of $\mathfrak{mod}(A_{1,m})$

To compute the presentation of $\mathfrak{mod}(A_{1,m})$ for a general m, things becomes a bit more complicated. Indeed we get a lot of representatives for the two cells and one cells while computing Brown's algorithm with the action of $\mathfrak{mod}(A_{1,m})$ on $\mathscr{SC}^m(A_{1,m})$. Nevertheless, using Theorem 4.1 in [Bro84], we obtain that $\mathfrak{mod}(A_{1,m})$ is finitely presented.

Theorem 5.22 (Theorem 4.1 in [Bro84])**.** *Let G be a group which acts on a simply-connected complex X such that:*

1. *Each vertex isotropy group is finitely presented.*
2. *Each edge isotropy group is finitely generated.*
3. *X has a finite 2-skeleton modulo the action by G.*

Then G is finitely presented.

Proposition 5.23. *Let $m \geq 3$, then $\mathfrak{mod}(A_{1,m})$ is finitely presented.*

Remark 5.24. In the case $m = 2$ one obtains $\mathcal{B}_2 \rtimes \mathbb{Z}$.

Proof. The group $G := \mathfrak{mod}(A_{1,m})$ acts on $\mathscr{SC}^m(A_{1,m})$, which is simply-connected by Proposition 5.10. The strategy is to apply the previous theorem. Let v be a vertex of height k in $\mathscr{SC}^m(A_{1,m})$.

Claim 5.25. There is an exact sequence:

$$1 \longrightarrow \mathcal{B}_k \xrightarrow{i} G_v \xrightarrow{\phi} \mathbb{Z}/m\mathbb{Z} \longrightarrow 1.$$

Proof. Let ϕ be the surjective morphism which associates to each element $g \in G_v$ its underlying permutation on the strands of $A_{1,m}$ at infinity. By definition, the permutation preserves the cyclic order of the m strands. Recall that the elements of G_v are the ones fixing the admissible surface associated to v in $\mathscr{SC}^m(A_{1,m})$ and being rigid outside it. Then the kernel of ϕ are the braided elements permuting the k punctures of the latter surface and being the identity outside, which give the braid group on k strands. □

Now, finite presentability of groups is closed under extension thus G_v is finitely presented. Then, one observes that given an edge e in $\mathscr{SC}^m(A_{1,m})$ from $o(e) = [\Sigma, \mathrm{id}]$ to $t(e) = [\Sigma \cup H, \mathrm{id}]$, where Σ has height k,

$$\begin{aligned} G_e &= \{g \in \mathfrak{mod}(A_{1,m}) \mid g\Sigma = \Sigma,\ g([\Sigma \cup H, \mathrm{id}]) = [\Sigma \cup H, \mathrm{id}], \text{ g rigid outside } \Sigma\} \\ &= G_{o(e)} \cap G_{t(e)} \cong \mathcal{B}_k, \end{aligned}$$

which is finitely generated.

Finally, it remains to show that the 2-skeleton of $\mathscr{SC}^m(A_{1,m})$ modulo the action by $\mathfrak{mod}(A_{1,m})$ is finite. Let Σ be an admissible surface and H_i, H_j two adjacent polygons of Σ in the ith (resp. jth) strand R_i (resp. R_j). Let $\epsilon^{\Sigma}_{i,j}$ be the minimum number of switch to send H_i to H_j via the clockwise or counterclockwise cyclic rotation around the central polygon (as β in Figure 45).

Claim 5.26. Let Σ and Δ be two admissible surfaces with the same height in $\mathscr{SC}^m(A_{1,m})$. Two 2-cells of the form ($[\Sigma, \mathrm{id}]$ to $[\Sigma \cup H_i \cup H_j, \mathrm{id}]$) and ($[\Delta, \mathrm{id}]$ to $[\Delta \cup J_k \cup J_l, \mathrm{id}]$) are in the same $\mathfrak{mod}(A_{1,m})$-orbit if and only if $\epsilon^{\Sigma}_{i,j} = \epsilon^{\Delta}_{k,l}$.

Proof. Assume that $\epsilon^{\Sigma}_{i,j} = \epsilon^{\Delta}_{k,l}$. Since Σ and Δ have the same height there is a rigid element from $\mathfrak{mod}(A_{1,m})$ mapping one of the two surfaces to the other (recall that we work with the spine). Then, since $\epsilon^{\Sigma}_{i,j} = \epsilon^{\Delta}_{k,l}$, there exists $p \geq 1$, β^p such that the $H_{i,j}$ are mapped to the $J_{k,l}$.

The converse follows from the fact that if the 2-cells are in the same $\mathfrak{mod}(A_{1,m})$-orbit, then there exists an element $g \in \mathfrak{mod}(A_{1,m})$ mapping Σ to Δ, the $H_{i,j}$ to the $J_{k,l}$ and rigid outside Σ. Hence, because of the rigidity we must have the same number of frontier arcs in Σ between H_i and H_j than in Δ between J_k and J_l (only cyclic rotations). □

Now since we are working with $\mathscr{SC}^m(A_{1,m})$ the height of the vertices is bounded. The previous claims tells us that two 2-cells with a root of height k are in the same $\mathfrak{mod}(A_{1,m})$ orbit if their corresponding $\epsilon_{i,j}$ are identical. Since we work with $A_{1,m}$, we have only a finite number of $\epsilon_{i,j}$. Hence, there is only a finite number of representatives for the 2-cells in $\mathscr{SC}^m(A_{1,m})$ modulo the action by $\mathfrak{mod}(A_{1,m})$. □

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